

Practice Guideline



AASLD Practice Guideline on blood-based noninvasive liver disease assessment of hepatic fibrosis and steatosis

PURPOSE AND SCOPE

Chronic liver disease (CLD) leads to liver fibrosis; it is associated with approximately two million annual deaths worldwide and is an enormous health burden.^[1,2] The majority of liver-related outcomes such as hepatic decompensation and complications from portal hypertension (variceal bleeding, hepatic encephalopathy, and ascites) and hepatocellular carcinoma (HCC) occur almost exclusively in those with advanced fibrosis. Therefore, it is critical to identify patients with any fibrosis and, in particular, moderate-to-advanced fibrosis. Over the past few decades, multiple noninvasive blood biomarkers and imaging modalities or tests, termed here “noninvasive liver disease assessment(s) (NILDA),” have been developed to determine the presence and severity of liver fibrosis (F), steatosis (S), and clinically significant portal hypertension.

NILDA can be generally categorized as blood-based and

imaging-based. The American Association for the Study of Liver Diseases (AASLD) Practice Guidelines Committee commissioned a diverse group of experts across multiple disciplines in the field of adult and pediatric liver disease to develop guidelines and guidance statements along with a systematic review covering blood-based NILDA to answer specific clinically focused questions (“patient, intervention, comparison, and outcome,” henceforth, PICO) (Table 1). This document focuses on the use of blood-based NILDA. The use of imaging-based NILDA^[3,4] in clinical practice and the use of blood and or imaging-based NILDA for assessment of clinically significant portal hypertension^[5,6] have been discussed elsewhere. These guidelines are intended primarily for adult and pediatric health care providers who see patients with CLD to provide a guidance algorithm that is summarized at the end of this document.

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Abbreviations: AASLD, American Association for the Study of Liver Diseases; ALD, alcohol-associated liver disease; ALT, alanine aminotransferase; APRI, AST-to-platelet ratio index; AST, aspartate aminotransferase; AT, ActiTest; AUROC, area under receiver operator curve; BA, biliary atresia; BARD, body mass index; AST/ALT ratio, and presence of type 2 diabetes mellitus; BMI, body mass index; CAP, Controlled attenuation parameter; CF, cystic fibrosis; CFLD, cystic fibrosis liver disease; CLD, chronic liver disease; CRN, clinical research network; CSPH, clinically significant portal hypertension; DAA, direct acting antiviral; DM, diabetes mellitus; DOR, diagnostic odds ratio; ELF, enhanced liver fibrosis; F, fibrosis (used in staging fibrosis with stages F1 to F4); FIB-4, Fibrosis-4 index; FLI, fatty liver index; GGT, gamma glutamyl transferase; GRADE, Grading of Recommendation Assessment, Development and Evaluation; HBeAg, hepatitis B envelope or “early” antigen; HBV, hepatitis B virus; HCC, hepatocellular carcinoma; HCV, hepatitis C virus; HIV, human immunodeficiency virus; HSI, hepatic steatosis index; HVPG, hepatic vein pressure gradient; IFN, interferon; LAP, lipid accumulation product; LR, likelihood ratio; LSM, liver stiffness measurement; MASLD, metabolic dysfunction-associated steatotic liver disease; METAVIR, meta-analysis of histological data in viral hepatitis; MRE, magnetic resonance elastography; MRI, magnetic resonance imaging; MRI-PDFF, magnetic resonance imaging-proton density fat fraction; NAFLD, nonalcoholic fatty liver disease; NAS, nonalcoholic fatty liver disease activity score; NASH, nonalcoholic steatohepatitis; NFS, nonalcoholic fatty liver disease fibrosis score; NILDA, noninvasive liver disease assessments; NPV, negative predictive value; PBC, primary biliary cholangitis; PICO, patient, intervention, comparison and outcome; PPV, positive predictive value; PRO-C3, N-terminal propeptide of type III collagen; PSC, primary sclerosing cholangitis; PT, prothrombin time; ROC, receiver operating characteristic curve; S, steatosis (used in staging steatosis with stages of 0–3); SLD, steatotic liver disease; SVR, sustained virologic response; SWE, Shear-wave elastography; T2DM, type 2 diabetes mellitus; TE, transient elastography; TG, triglycerides; US, ultrasound; WC, waist circumference; α 1AT, alpha-1-antitrypsin.

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METHODS

Overall approach

The guideline writing group consisted of a multidisciplinary panel of experts in both adult and pediatric hepatology, pathology, and radiology, including methodology experts. Two complementary approaches were taken to answer the PICO questions relevant to various CLDs. Autoimmune hepatitis (AIH) has been reviewed and discussed elsewhere.^[7] The first approach depended on a commissioned systematic review conducted independently by the Mayo Clinic

Evidence-Based Practice Center (Supplemental Figure S1, <http://links.lww.com/HEP/I455>); this led to graded recommendations following the Grading of Recommendations, Assessment Development, and Evaluation system (GRADE) approach (Table 2).^[8,9] These recommendations are followed by a section that describes the quality of evidence, when applicable, and other considerations. The panelists monitored the literature for studies published during the systematic review's search date and included relevant studies through April 2022. Strength of recommendations was based on the quality of the evidence, balance of benefits and harms, the burden of testing (access and financial), and feasibility of the recommended action. The "strength of recommendation" determination assumed that performing tests with acceptable (> 70%), excellent (> 80%), or outstanding (> 90%) diagnostic accuracy are associated with improved patient outcomes. The recommendations were graded as either strong (apply to most patients with minimal variation and can be adapted as policy in most situations) or conditional (apply to a majority of patients, but variation in care is acceptable). These recommendations are followed by a section that describes the quality of evidence (if applicable) and other considerations. Because of the rapid evolution of the field and predetermined quality of studies incorporated in our systematic reviews, we were not able to include every published study on the topic. In particular, studies with smaller sample sizes (< 50 individuals) or those with mixed etiology were excluded.

TABLE 1 PICO questions in NILDA

Blood-based testing for fibrosis or steatosis in adults

PICO 1	In adult patients with chronic liver disease, including hepatocellular (HCV, HIV-HCV, HBV, HCV/HBV, HIV/HBV, NAFLD, and ALD) or cholestatic (PSC and PBC) disorders, are blood-based biomarker panels accurate in staging hepatic fibrosis (F0-1 vs. F2-4, F0-2 vs. F3-4, and F0-3 vs. F4) using histopathology as the reference?
PICO 2	In adult patients with chronic liver disease, including hepatocellular (HCV, HIV-HCV, HBV, HIV-HBV, NAFLD, and ALD) or cholestatic (PSC and PBC) disorders, is any blood-based biomarker panel superior to another blood-based biomarker panel in staging hepatic fibrosis (F0-1 vs. F2-4, F0-2 vs. F3-4, and F0-3 vs. F4) using histopathology as the reference?
PICO 3	In adult patients with chronic liver disease, including hepatocellular (HCV, HIV-HCV, HBV, HIV-HBV, NAFLD, and ALD) or cholestatic (PSC and PBC) disorders, is the combination of two blood-based biomarker panels superior to a single one for staging fibrosis (F0-1 vs. F2-4, F0-2 vs. F3-4, and F0-3 vs. F4) using histopathology as the reference?
PICO 4	In adult patients with chronic liver disease, including hepatocellular (HCV, HIV-HCV, HBV, HIV-HBV, NAFLD, and ALD) or cholestatic (PSC and PBC) disorders, do serial blood-based biomarker panels accurately predict the natural history of progression of fibrosis or regression of fibrosis in response to therapy relative to serial histopathology as the reference?
PICO 5	In patients with NAFLD, are blood-based biomarker panels accurate in grading hepatic steatosis (S0 vs. S1-3, S0-1 vs. S2-3, and S0-2 vs. S3) using histopathology or MR-spectroscopy or MRI PDFF as the reference?

Blood-based testing in children

PICO 6	In pediatric chronic liver disease (HCV, HBV, BA, CFLD, and NAFLD/NASH), are blood-based biomarkers accurate in staging hepatic fibrosis (F0-1 vs. F2-4, F0-2 vs. F3-4, and F0-3 vs. F4) using histopathology as the reference?
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ALD, alcohol-associated liver disease; BA, biliary atresia; CFLD, cystic fibrosis liver disease; F, fibrosis; HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus; MR, magnetic resonance; MRI PDFF, magnetic resonance imaging proton density fat fraction; NAFLD, nonalcoholic fatty liver disease; NASH, nonalcoholic steatohepatitis; PBC, primary biliary cholangitis; PICO, Patient, Intervention, Comparison and Outcome; PSC, primary sclerosing cholangitis.

TABLE 2 GRADE approach^a

1. Rating the quality of evidence			
Study design:	Initial rating of quality of evidence:	Rate down when:	Rate up when:
RCT	High	Risk of bias	Large effect size (e.g., RR = 0.5)
Observational	Moderate	Inconsistency	Very large effect (e.g., RR = 0.2)
	Low	Imprecision	Dose-response gradient
	Very low	Indirectness	All plausible confounding would increase the association
		Publication bias	
2. Determinants of strength of a recommendation			
Quality of evidence			
Balance of benefits and harms			
Patient values and preferences			
Resources and costs			
3. Implications of the strength of a recommendation			
Strong			
Population: Most people in this situation would want the recommended course of action, and only a small proportion would not.			
Health care workers: Most people should receive the recommended course of action.			
Policy makers: The recommendation can be adopted as policy in most situations.			
Conditional			
Population: The majority of people in this situation would want the recommended course of action, but many would not.			
Health care workers: Be prepared to help patients make a decision that is consistent with their values using decision aids and shared decision-making.			
Policy makers: There is a need for substantial debate and involvement of stakeholders.			

^aModified from references.^[8,9]
Abbreviations: GRADE, Grading of Recommendations, Assessment Development, and Evaluation system; RCT, randomized controlled trial; RR, relative risk.

In order to address several other important clinical questions that could not be answered by a systematic review due to sparse and/or indirect evidence, the second approach involved a thorough narrative review by the writing group to develop ungraded guideline statements. These ungraded statements considered additional sources and the clinical experience of the authors with regard to noninvasive assessments of hepatic fibrosis and steatosis. Technical remarks and supporting evidence for graded and ungraded statements are included with recommendations to help reconcile the level of the recommendation with the quality of the evidence and to facilitate implementation. For these guideline statements (below) on blood-based NILDA, adults are defined as being at least 18 years of age, and pediatrics are younger than age 18 years.

Consensus process

For all guideline statements, we pursued a consensus approach to define the final set of recommendations using previously described methodology and also adapted by the AASLD practice metrics committee.^[10] Statements with <75% agreement were rediscussed with the following: 1) review of the scores; 2) discussion to identify the reasons for variation; 3) revision of suboptimally worded statements for accuracy by consensus; 4) deletion of statements that were deemed

problematic or irrelevant by consensus; and 5) identification of additional statements deemed necessary for inclusion in the list of statements.

Rationale for NILDA

Accurate assessment of the degree of liver fibrosis and steatosis is essential in predicting prognosis and making treatment recommendations in patients with CLD. Although liver biopsy has long been the reference standard for assessing fibrosis and steatosis, it is costly, invasive, and carries a small, but important, risk of complications.^[11,12] Pain is the most common, whereas clinically apparent bleeding occurs in some one in every five hundred liver biopsies (rate of 0.2%), with severe bleeding in one out of every two thousand five hundred to one in ten thousand (rate of 0.04% to 0.01%).^[13] The mortality rate associated with liver biopsy is estimated to be one per ten thousand to one per twelve thousand (rate of 0.01% to 0.0083%).^[11] Biopsy complication rate varies based on operator experience, underlying comorbidities, size of the needle, number of passes, and underlying bleeding risk due to low platelets and/or increased prothrombin time.

Current noninvasive assessments rely on biochemical (blood) or physical (imaging) characteristics that are developed in relation to cross-sectional, histopathologic scores and do not account for the dynamic progression of

fibrogenesis or variable disease etiology pathogenesis. In the last 20 years, noninvasive methods for assessing liver fibrosis and steatosis utilizing blood- and imaging-based methods have been developed to reduce the need for invasive liver assessment procedures.

Histopathological principles underlying NILDA

Fibrosis scores are generally disease-specific and technically cannot be unified across different CLDs. To achieve a cohesive approach for the purposes of NILDA, the writing group incorporated the various fibrosis staging systems into a single one and classified them into at least significant fibrosis (equivalent to at least fibrosis stage 2 or F2-4), at least advanced fibrosis (F3-4), and cirrhosis (F4). For simplicity, the Guidelines statements employ the generic “F” stages throughout the text. Various histologic scoring systems to stage fibrosis and grade inflammation and steatosis have been used as standard reference measures in studies validating NILDA biomarkers (Table 3).^[14–22]

Although differences are subtle in most instances among different liver histologic scoring schemes for fibrosis, using scores interchangeably between and among different schemes is problematic (Table 3). For example, Scheuer stage 3 is not equivalent to the meta-analysis of histological data in viral hepatitis (METAVIR) F3. The Ishak system has seven possible scores,^[23–25] which allows for finer detail in fibrosis scoring; a challenge lies with scores five and six in that most treating physicians assume that score five is cirrhosis based on prognostic implications.^[26] However, because Ishak 5 is defined as “marked bridging with occasional nodules” or “incomplete cirrhosis,” and the definition of cirrhosis is diffuse parenchymal nodularity; Ishak 5 does not meet these criteria.^[27] In adult patients with fatty liver disease, whether alcohol-associated or due to metabolic syndrome, fibrosis initially occurs in zone 3 (centrilobular area) with a perisinusoidal and pericellular pattern. In contrast, fibrosis in other types of CLD is largely portal-based. In children, fibrosis is often triggered by a genetic or persistent environmental insult or by biliary injury with duct obstruction. Thus, the patterns of fibrosis distribution depend on the etiology, susceptibility, and response to injury.

We acknowledge that there has been a recent multisociety endorsement of a nomenclature change from NAFLD to metabolic dysfunction–associated steatotic liver disease (MASLD). Although this is an important change that will impact of future of the study of this entity, all data utilized to develop these guideline statements were based on prior literature that utilized the previous NAFLD definition. Therefore, NAFLD is the term used throughout this document when referring to the existing literature. Current evidence indicates >98% overlap

between patients who meet criteria for diagnosis of NAFLD/NASH and the new criteria for MASLD/metabolic dysfunction–associated steatohepatitis (MASH) in large cohort studies, indicating that the analyses and recommendations provided in these Guidelines for patients with NAFLD/NASH are likely to pertain to patients characterized by the new nomenclature of MASLD and MASH.

The two most commonly used scoring systems in steatotic liver disease (SLD) for steatosis and fibrosis in NAFLD are those by Brunt and the NASH Clinical Research Network (CRN), i.e., the NAFLD Activity Score (NAS).^[21,22] The Brunt scoring system has four possible grades (0–3) and five possible stages (0–4). Both systems determine the degree of steatosis based on the percentage of steatotic hepatocytes involved: normal <5%, mild = 5% to 33%, moderate = 34% to 66%, and severe >66% (Table 3). In children with NASH, steatosis is more profound, and the distribution of fibrosis and inflammation is found primarily and initially in zone 1 (periportal).^[28]

Some experts have suggested that the grading and staging of NAFLD may also be applied to alcohol-associated liver disease (ALD) due to similarity and overlap in morphological features.^[29] Histologic scoring systems specifically for ALD have been proposed over the years,^[30,31] but none have been used in standard clinical practice. One scoring system has been proposed for alcoholic hepatitis, which correlates histological features with prognosis.^[20] Although advanced fibrosis was identified as an independent predictor of short-term mortality, i.e., indicating chronicity and progression of disease, this was not the main outcome of the study; therefore, this histologic scoring system has not been applied in clinical practice.^[20] Additionally, liver biopsies may not be routinely obtained in patients with suspected ALD, leading to challenges in correlating liver histology with outcome.

Although liver histology is considered the reference standard to which NILDA is assessed, several factors can bias liver histology, including sampling bias, classification bias, and spectrum bias. Liver biopsy specimen size and adequate number of portal tracts are very important to reduce sampling bias.^[11,32,33] Unfortunately, most published studies have not adjusted for this bias.^[34,35] Quantitative techniques such as histomorphometry using collagen- or fat-specific stains have been introduced to overcome inherent problems encountered in semiquantitative histological staging systems.

Evidence using NILDA has suggested that fibrosis can regress (suggesting that the total amount of fibrosis in the liver becomes reduced; this does not, however, necessarily mean that the liver architecture becomes normal), particularly once the cause of liver injury is resolved.^[36,37] Unfortunately, there is no histopathological score that has been validated for use in regression of fibrosis, despite reports characterizing regression of fibrosis features,

TABLE 3 Staging of fibrosis across multiple liver diseases and corresponding classification scores

a. Staging of Fibrosis Across Multiple Liver Diseases and Corresponding Classification Scores					
	Fibrosis stage				
	0	F1	F2	Significant fibrosis	
				Advanced fibrosis	
				F3	Cirrhosis F4
Scheuer/Batts-Ludwig (Viral and autoimmune hepatitis) ^[14,15]	No fibrosis	Enlarged, fibrotic portal tracts	Periportal or P-P septa but intact architecture	Fibrosis with architectural distortion but no obvious cirrhosis	Probable or definite cirrhosis
Knodell (Viral and autoimmune hepatitis) ^[16]	No fibrosis	Fibrous portal expansion	N/A	Bridging fibrosis	Cirrhosis
Ishak (Various etiologies) ^[17]	0: No fibrosis	1: Fibrous expansion of some portal areas, with or without short fibrous septa	2: Fibrous expansion of most portal areas, with or without short fibrous septa 3: Fibrous expansion of most portal areas with occasional portal to portal bridging	4: Fibrous expansion of portal areas with marked bridging 5: Marked bridging (P-P and/or P-C) with occasional nodules (incomplete cirrhosis)	6: Cirrhosis (probable or definite)
Meta-analysis of histologic data in viral hepatitis (METAVIR) (Various etiologies) ^[18]	No fibrosis	Stellate enlargement of portal tract but without septa formation	Enlargement of portal tract with rare septa formation	Numerous septa without cirrhosis	Cirrhosis
Ludwig (PBC and PSC) ^[19]	N/A	N/A	N/A	Bridging fibrosis	Cirrhosis
Alcohol-associated liver disease (alcohol hepatitis histological score) ^[20]	No fibrosis or portal fibrosis	Expansive periportal fibrosis	Bridging fibrosis	Cirrhosis	
Brunt-Kleiner (NAFLD) ^[21,22]	No fibrosis	1°: Delicate perisinusoidal 1B: Dense perisinusoidal 1C: portal-only fibrosis	Perisinusoidal and portal/periportal fibrosis	Bridging fibrosis	Cirrhosis

TABLE 3. (continued)

a. Staging of Fibrosis Across Multiple Liver Diseases and Corresponding Classification Scores				
Fibrosis stage				
Significant fibrosis				
Advanced fibrosis				
Cirrhosis				
F4				
F3				
F2				
F1				
0				
b. Assessment and Grading of Steatosis Based On the Percent of Hepatocytes Affected				
Degree of steatosis				
0 (Normal or minimal)	1 (Mild)	2 (Moderate)	3 (Severe)	
< 5%	5%–33%	34%–66%	> 66%	

Based on references Kleiner et al. [21] and Brunt et al. [22]
Abbreviations: N/A, not applicable; NAFLD, nonalcoholic fatty liver disease; PBC, primary biliary cholangitis; P-C, port-central; P-P, portal-portal; PSC, primary sclerosing cholangitis.

such as thinning and perforation of septa, isolated collagen fibers not attached to a portal tract/central vein, and changes in baseline architectural distortion, including loss of zonation of vascular structures.^[38,39]

Assessment of diagnostic performance of noninvasive markers

We used several statistical tests and indices in our assessment of the performance of blood-based NILDA (Table 4). Although several studies have reported test characteristics such as sensitivity and specificity at a selected cutoff, the positive and negative predictive values of the test are dependent on the prevalence of the condition (e.g., fibrosis or steatosis).^[40] The Likelihood Ratio (LR) is defined as the likelihood that a test result would be expected if the patient had the disease compared with the likelihood of this same result in a patient without the disease. Positive LR describes the odds of having fibrosis or steatosis among patients with a positive test, whereas negative LR describes the odds of having fibrosis or steatosis in patients with a negative test. Positive LR above 10 and negative LR below 0.1 suggest strong diagnostic evidence. The diagnostic odds ratio (DOR) is the ratio of the odds of disease in those who test positive to the odds of the disease in those who test negative (i.e., summarizing the odds of fibrosis in those with a positive test relative to those with a negative test) and provides a reliable estimate of a test’s accuracy that is independent of the prevalence of the condition being tested. The area under the receiver operating characteristic curve (AUROC) analysis is another effective way to summarize the overall diagnostic accuracy of the test. The AUROC ranges from 0 to 1, where a value of 0 indicates a perfectly inaccurate test, and a value of 1 reflects a perfectly accurate test. In general, an AUROC of 0.5 suggests no discrimination (i.e., inability to diagnose patients with and without the disease or condition based on the test), 0.7 to 0.8 is considered acceptable, 0.8 to 0.9 is considered good, and more than 0.9 is considered excellent.

Blood-based biomarkers

Blood-based assessment of fibrosis takes advantage of the complex and dynamic interplay between the inflammatory response and fibrogenesis, including elements of extracellular matrix synthesis and degradation. Noninvasive blood-based biomarkers include combinations of tests of “direct” markers, which are mostly complex macromolecules derived from myofibroblasts and extracellular matrix remodeling, or “indirect” markers reflective of inflammation and/or portal hypertension. Although blood-based tests were initially developed for hepatitis C virus (HCV), many

TABLE 4 Diagnostic performance indices used in NILDA

Diagnostic index	Calculation	Comments
Sensitivity	$TP/(TP + FN)$	Not dependent on the prevalence of the condition in the population. High sensitivity helps rule out the disease (few FNs).
Specificity	$TN/(TN + FP)$	Not dependent on the prevalence of the condition in the population. High specificity helps ruling in disease (few FPs).
Accuracy	$(TP + TN)/(P + N)$	
PPV	$TP/(TP + FP)$	The probability that a person with a positive test indeed has the disease or condition of interest Affected by the prevalence of the disease in the population.
NPV	$TN/(TN + FN)$	The probability that a person with a negative test does NOT have the disease or condition of interest. Affected by the prevalence of the disease in the population.
Positive LR	$\text{Sensitivity}/(1-\text{Specificity})$ OR TP/P	Positive LR greater than 10 suggests strong test to predict outcome.
Negative LR	$(1-\text{Sensitivity})/\text{Specificity}$ OR TN/N	Negative LR less than 0.1 suggests strong diagnostic evidence for not having the outcome.
DOR	Positive LR/Negative LR	The ratio of odds of positivity of those with disease relative to odds of positivity in those without disease. The higher the DOR, the better the test.
AUROC	Graph values of test performance from 0 (a perfectly inaccurate test) to 1 (a perfect test). Plots the diagnostic ability of a binary classifier system as its discrimination threshold is varied.	Summarizes the overall diagnostic accuracy of a test. In general, an AUROC of 0.5 suggests no discrimination (i.e., ability to diagnose patients with and without the disease or condition based on the test), 0.7 to 0.8 is considered acceptable, 0.8 to 0.9 is considered excellent, and more than 0.9 is considered outstanding.

Abbreviations: AUROC, area under the receiver operating characteristic curve; DOR, diagnostic odds ratio; FN, false-negative; FP, false-positive; LR, likelihood ratio; N, all negative tests; NILDA, noninvasive liver disease assessments; NPV, negative predictive value; P, all positive tests; PPV, positive predictive value; TP, true positive; TN, true negative.

have been adopted to assess fibrosis in other CLDs, including NAFLD. Algorithms used are conceptually divided into the following: 1) simple, nonproprietary models that include routine blood tests; 2) those that combine routine tests with clinical variables; and 3) more complex proprietary models that include direct measurements of collagen synthesis or degradation with or without clinical variables (Table 5).^[41–51]

Commonly used clinical variables are age, sex, body mass index (BMI), and the presence of diabetes mellitus (DM). Complex models include direct measurements of collagen synthesis and degradation (hyaluronic acid, N-terminal propeptide of type III procollagen, matrix metalloproteinase type 1 and 2, tissue inhibitors of matrix metalloproteinases type 1 and 2, α 2-macroglobulin, apolipoprotein A1, transforming growth factor- β 1, procollagen type 1 carboxy-terminal peptide, chitinase-3-like protein 1 [YKL-40], and/or cytokeratin-18 fragments).^[41–43,45–50,52] However, blood-based tests may be limited by clinical

factors such as systemic inflammation or sepsis (Table 6).^[53–62]

Unreliable classifications for blood-based biomarker algorithms that utilize bilirubin may occur in hemolysis, Gilbert's syndrome, or cholestasis. Other clinical disease states such as acute hepatitis, sepsis, and systemic inflammatory conditions may produce false-positive results in blood biomarker algorithms that incorporate amino-transferases or acute phase reactants such as hyaluronic acid, α -2 macroglobulin, platelets, N-terminal propeptide of procollagen type III, or false-negative results with elevated haptoglobin. Simple markers may have lower accuracy for advanced fibrosis in patients with HCV with end-stage renal disease and normal-range transaminases.^[58] Hyaluronic acid levels may be influenced by age^[63] or postprandial state.^[59,64] HIV co-infection may result in thrombocytopenia or may be associated with drug-induced elevations in bilirubin or γ -glutamyl transferase (GGT), which can also affect diagnostic accuracy of several blood-based marker panels.

TABLE 5 Components of blood-based biomarker algorithms for fibrosis^a

Blood-marker panel, year (reference)	Disease cohort	Clinical variables	Indirect markers	Direct markers	Model algorithm
Simple blood-based NILDA with or without clinical data					
APRI, 2003 ^[41]	HCV	—	AST, platelets	—	$[(\text{AST level}/\text{ULN})/\text{platelet count (10}^9/\text{L)}] \times 100$
FIB-4, 2006 ^[42]	HIV-HCV	Age	AST, ALT, platelets	—	$\text{age (y)} \times \text{AST (U/L)}/\text{platelet count (10}^9/\text{L)} \times \sqrt{\text{ALT (U/L)}}$
NFS, 2007 ^[43]	NAFLD	Age, BMI, IFG/diabetes	AST, ALT, platelets, albumin	—	$-1.675 + (0.037 \times \text{age}) + (0.094 \times \text{BMI}) + 1.13 \times \text{IFG/diabetes (yes = 1, no = 0)} + 0.99 \times (\text{AST/ALT ratio}) - (0.013 \times \text{platelets}) - (0.66 \times \text{albumin})$
Fibroindex (2007) ^[44]	HCV	—	AST, platelets, gamma globulin	—	$1.738 - 0.064(\text{platelet } [\times 10^4/\text{mm}^3]) + 0/005(\text{AST IU/L}) + 0.463(\text{gamma globulin [g/Dl]})$
King's Score, 2009 ^[45]	HCV	Age	AST, INR, platelets	—	$\text{Age} \times \text{AST} \times \text{INR}/[\text{platelet count (109/L)}]$
Easy Liver Fibrosis Test (Elift), 2017 ^[46]	Mixed	Age, sex	GGT, AST, platelets, Prothrombin Index	—	Component weighted scores (0-4)
Complex, proprietary blood-based NILDA					
FibroSure™/FibroTest®, 2001 ^[47]	HCV	—	$\alpha 2\text{M}$, GGT, total bilirubin, haptoglobin, ApoA-I ¹	—	Proprietary
ELF™, 2004 ^[48]	Mixed	Age	—	HA, PIIINP, TIMP-1	Proprietary
FibroSpect II™, 2004 ^[49]	HCV	—	$\alpha 2\text{M}$	HA, TIMP-1	Proprietary
HepaScore™, 2005 ^[50]	HCV	Age, sex	Total bilirubin, $\alpha 2\text{M}$, GGT	HA	Proprietary
FibroMeter™, 2005 ^[51]	Mixed	Age	Platelets, Prothrombin Index, urea, AST, $\alpha 2\text{M}$	HA	Proprietary

Abbreviations: ALT, alanine aminotransferase; ApoA-1, apolipoprotein A-1; APRI, AST-to-platelet Ratio Index; AST, aspartate aminotransferase; BMI, body mass index; ELF, enhanced liver fibrosis; FIB-4, Fibrosis-4 index; GGT, gamma-glutamyl transferase; HA, hyaluronic acid; IFG, impaired fasting glucose; INR, international normalized ratio (also known as prothrombin time); L, liter; NAFLD, nonalcoholic fatty liver disease; NFS, NAFLD fibrosis score; PIIINP, amino-terminal propeptide of type III procollagen; PT, prothrombin time; TIMP-1, tissue inhibitor matrix metalloproteinase 1; U, units; ULN, upper limit of normal common blood tests (includes the following: AST, ALT, platelet count, albumin, gamma-globulin, GGT, haptoglobin, PT, and total cholesterol); A2M, $\alpha 2$ -macroglobulin.

^aOriginal study cohorts are referenced.

RECOMMENDATIONS AND GUIDELINE STATEMENTS

PICO 1: In adult patients with chronic liver disease, including hepatocellular (HCV, HIV-HCV, hepatitis B virus [HBV], HIV-HBV, NAFLD, and ALD) or cholestatic (primary sclerosing cholangitis [PSC] and primary biliary cholangitis [PBC]) disorders, are blood-based biomarker panels accurate in staging hepatic fibrosis (F0-1 vs. F2-4, F0-2 vs. F3-4, and F0-3 vs. F4) using histopathology as the reference?

Guideline Statements

1. In adult patients with chronic HBV and HCV undergoing fibrosis staging prior to antiviral therapy, AASLD recommends using simple blood-based NILDA such as APRI or Fibrosis-4 Index (FIB-4) as an initial test to detect significant (F2-4), advanced fibrosis (F3-4) or cirrhosis (F4) compared with no test (strong recommendation, moderate quality of evidence).
2. In adult patients with NAFLD undergoing fibrosis staging, AASLD recommends using simple blood-based NILDA tests such as FIB-4 to detect advanced fibrosis (F3-4) compared to no test (strong recommendation, moderate quality of evidence).
3. In adult patients with ALD or chronic cholestatic liver disease undergoing fibrosis staging, there is insufficient evidence to recommend using blood-based NILDA for staging fibrosis (ungraded statement).

TECHNICAL REMARKS

- Direct and indirect blood biomarkers include components (bilirubin, aminotransferases, platelets, and other acute-phase reactants) that may be associated with false-positive or false-negative test results in patients with certain disorders such as acute hepatitis, hemolysis, Gilbert's syndrome, human immunodeficiency virus (HIV)-induced thrombocytopenia, splenectomy, and disease or treatment-related elevation in bilirubin or aminotransferases (Table 6).
- Blood-based biomarkers have high sensitivity and negative predictive value (NPV) for "ruling out" advanced fibrosis in NAFLD but low positive

predictive value (PPV) to "rule-in" in advanced fibrosis in low prevalence cohorts (supplemental Table S1, <http://links.lww.com/HEP/I343>, Figure 1, Table 7).^[43,54,65–90]

- There are no validated blood-based biomarker thresholds that correlate with the fibrosis stage following sustained virologic response (SVR) in patients with HCV. Both indirect and direct blood biomarkers are associated with high false-negative rates for advanced fibrosis following antiviral therapy in patients with HBV or HCV.
- Although not included in the systematic review, NFS can be used to detect F3-4 in those with NAFLD.

BACKGROUND

Although none of the current blood-based biomarkers are liver-specific, potential advantages include availability (for simple nonproprietary tests), interlaboratory reproducibility, and ease of use in routine clinical practice. However, an important consideration is the reliability of currently available blood-based markers to classify patients with CLD accurately. For example, prior modeling in HCV has indicated that because of sampling error, liver histology (the reference standard to which NILDA are compared with) is imperfect; therefore, the ideal biomarker performance usually does not exceed an AUROC of 0.9.^[91] However, these performance measures do not overcome limitations related to disease heterogeneity and spectrum effect/bias in study cohorts.^[92]

EVIDENCE AND RATIONALE

HCV

In the current era of direct-acting antiviral (DAA) therapies with high efficacy for HCV, excluding stage F0-1 prior to treatment is less clinically relevant than the detection of advanced fibrosis (F3-4) or cirrhosis (patients with advanced disease should have ongoing post-treatment HCC surveillance). A systematic review of 10 different simple and complex biomarker panels concluded that clinically relevant predictive values (PPV $\geq$ 90% and NPV $\geq$ 95%) for significant fibrosis (F2-4) could be obtained for only 35% of patients with HCV before therapy.^[67] Aspartate aminotransferase (AST)-to-platelet ratio index (APRI) and FIB-4 are the best validated of the simple, cheap, and readily available nonproprietary tests, but they are known to be associated with "indeterminate" range scores and unreliable diagnostic performance in some patients. FibroTest™ (Bio-Predictive, Paris, France) or in the United States, FibroSURE® (LabCorp, Burlington, North Carolina) are the most validated blood-based biomarkers with a proprietary algorithm. A meta-analysis of 172 studies

TABLE 6 Clinical factors affecting performance of blood-based noninvasive assessment of fibrosis

Clinical condition	Tools affected	Comments
Age	FIB-4 NFS King's eLift ELF™ Hepascore™ FibroMeter™	In the age extremes (both very young and very old), may not perform as well.
Splenectomy	APRI FIB-4 Fibroindex FibroMeter™ NFS	Because these tools use platelets as a biomarker of portal hypertension, attenuated thrombocytopenia from splenectomy gives a falsely lower estimation.
Thrombocytopenia (not related to portal hypertension)	APRI FIB-4 Fibroindex FibroMeter™ NFS	Because these tools use platelets as a biomarker of portal hypertension, thrombocytopenia from other conditions gives a falsely higher estimation.
Active alcohol use ^[53]	FibroTest™ HepaScore™	Increases GGT, leading to falsely elevated estimation.
Elevated ALT and/or AST (inflammatory hepatitis) ^[53–55]	APRI FIB-4 Fibroindex FibroMeter™ NAFLD fibrosis score	Elevated aminotransferases occurring in relation to acute or acute-on-chronic hepatitis lead to falsely elevated estimation.
Chronic kidney disease ^[56–58]	Fibroindex APRI FIB-4 FibroMeter™	Elevated urea levels can result in falsely lower estimation. Hemodialysis patients tend to have lower ALT and AST levels, resulting in falsely lower estimation. Hemofiltration can result in lower stiffness in patients with baseline fluid overload.
Malnutrition	NAFLD fibrosis score	Albumin reduction that is disproportionate to liver dysfunction results in falsely elevated estimation.
Inflammatory condition	FibroTest™ Fibroindex HepaScore™ FibroMeter™	Can result in increased α 2-macroglobulin levels and falsely elevated Fibrotest, and increased α -globulin and falsely elevated Fibroindex.
Hemolysis	FibroTest™ Hepascore™	Decreases haptoglobin levels and increases total bilirubin leading to falsely elevated estimation.
Gilbert syndrome and other cholestatic diseases	FibroTest™ Hepascore™	Can result in increased total bilirubin and falsely elevated estimation.
Postprandial ^[59]	NFS	Liver stiffness increases up to 26% have been described for TE-LSM 2 h after a meal. A rise in postprandial glucose (> 110 mg/DI) falsely elevates NAFLD fibrosis score.
Gastrectomy ^[60]	FibroSpect™ HepaScore™ ELF™	Increases hyaluronic acid resulting in falsely elevated estimation.
Extra-hepatic fibrosing conditions ^[61]	FibroMeter™ FibroSpect™ ELF™	Conditions such as interstitial lung disease can increase collagen turnover markers resulting in elevated estimation.
Acute sickle cell crisis ^[62]	FibroTest™	Related to hemolysis (as aforementioned); Decreases haptoglobin levels and increases total bilirubin leading to falsely elevated estimation.

Abbreviations: ALT, alanine aminotransferase; APRI, AST-to-platelet ratio index; AST, aspartate aminotransferase; BMI, body mass index; ELF, enhanced liver fibrosis; FIB-4, Fibrosis-4 index; GGT, gamma glutamyl transferase; NAFLD, nonalcoholic fatty liver disease; NFS, nonalcoholic fatty liver disease fibrosis score.

evaluated several blood-based biomarkers in patients with HCV and indicated that blood-based NILDA tests had moderate diagnostic utility for the detection of F2-4 and F4.^[93] Our systematic review^[94] indicated that both simple and complex blood-based NILDA had acceptable diagnostic performance for detecting F2-4, F3-4, and F4 in patients with HCV prior to antiviral therapy (supplemental Table S1, <http://links.lww.com/HEP/I343>).

Liver biopsies are no longer performed routinely in patients with HCV who are post-SVR, and the diagnostic role of indirect and direct blood-based biomarkers for staging fibrosis in these patients has not been established. In general, routine use of blood-based biomarkers that include aminotransferases is likely to be associated with a high false-negative rate for advanced disease following viral clearance. A study in 115 patients with HCV and biopsy available 5-years post-SVR noted AUROC for APRI and FIB-4 of 0.81 to 0.88 for F2-4 and F3-4, although the selected biomarker thresholds were much lower post-SVR.^[95] A smaller study of 38 patients with HCV stage F4 and biopsy 5-years post-SVR also noted lower scores for both indirect (APRI, FIB-4, King's score) and direct (European Liver Fibrosis [ELF], Siemens Healthineers AG, Erlangen, Germany) biomarkers, with an AUROC of 0.58 to 0.63 for F4 post-SVR.^[96] Thus, validation of post-SVR biomarker thresholds that correspond to fibrosis stages is required.

HBV

Management decisions in HBV infection consider not only fibrosis stage but also disease activity based on HBV DNA levels, alanine aminotransferase (ALT) elevation, and HBe-antigen (HBeAg) status, along with other variables.^[97] Although blood-based biomarkers of fibrosis have not been routinely adopted for the management of HBV infection, detection of advanced fibrosis or cirrhosis has important prognostic implications. A meta-analysis of 30 studies with APRI, FIB-4, and FibroTest indicated a summary AUROC of 0.75 to 0.84 for F2-4 and 0.75 to 0.90 for F4.^[98] Another meta-analysis of 16 studies that included 2494 patients with HBV (including 1754 with F4) indicated summary AUROC for FibroTest of 0.84 for F2-4 and 0.87 for F4.^[99] Our systematic review,^[94] which included 96 studies, indicated that APRI and FIB-4 had acceptable diagnostic performance for F2-4, F3-4, and F4 in patients with HBV and higher specificity (>0.80) at upper test cutoffs. A study in 510 patients with HBV or HCV indicated that optimal sensitivity cutoffs for F3-4 and F4 using FibroTest, FibroMeter®, and HepaScore were lower in HBV compared with HCV. These findings suggest that the use of thresholds established in HCV can result in higher false-negative rates for advanced fibrosis and cirrhosis in HBV.^[100]

NAFLD

Increased fibrosis stage has important prognostic implications in NAFLD.^[101,102] Revised FIB-4 thresholds of ≤ 1.30 and ≥ 2.67 have been proposed as having higher predictive values for F3-4 in the NASH CRN cohort.^[103] However, a prior meta-analysis that included six studies with 1910 patients noted that FIB-4 ≥ 2.67 and ≥ 3.25 both had a summary specificity of 0.96 to rule-in advanced fibrosis.^[104] Our systematic review of 32 studies that reported these upper FIB-4 thresholds for NAFLD advanced fibrosis indicated similar pooled specificity of 0.94 for both FIB-4 ≥ 2.67 and ≥ 3.25 .^[94] Our results also indicated DOR of 7.81 and 10.19 for F3-4 at the lower FIB-4 thresholds of 1.3 and 1.45 and 10.76 and 7.01 for upper thresholds of 2.67 and 3.25, respectively. The NAFLD fibrosis score (NFS) was developed as a simple scoring algorithm to reduce the need for a liver biopsy to identify patients with NAFLD with advanced fibrosis.^[43] Optimal test thresholds for selecting F3-4 using blood-based markers vary between studies due to differences in population characteristics and disease prevalence compared with the original test derivation cohort.^[104] Our comprehensive review of NFS included 11,372 patients with NAFLD with advanced fibrosis on biopsy and assessed NFS performance at the original validated lower and upper thresholds of -1.455 and 0.676 , respectively. At advanced fibrosis prevalence rates that varied from 3% to 80%, the summary median (95% confidence interval [CI]) sensitivity for excluding F3-4 at less than -1.455 was 0.75 (95% CI: 0.61–0.81), and specificity for diagnosing F3-4 at greater than 0.676 was 0.96 (95% CI: 0.93–0.98), with indeterminate rates of 33.5% (95% CI: 25.6–44.4; Table 7).

This is comparable with an individual patient meta-analysis of 3248 patients with NAFLD that resulted in specificity of 0.91 for F3-4 at established cutoffs for NFS and indeterminate rates of 39%.^[105] Consideration of disease prevalence in the target population is important because many of these simple and proprietary blood-based markers will be increasingly used to screen for advanced fibrosis in lower prevalence nontertiary cohorts at risk of NASH. A meta-analysis of 11 studies using ELF tests for F3-4 noted a high sensitivity (0.93) but limited specificity (0.34) at the lower recommended threshold of 7.7; higher thresholds and F3-4 prevalence of at least 30% were required for increasing ELF PPV to >0.8 for advanced fibrosis.^[106] Overall, both simple and complex blood-based marker algorithms have acceptable diagnostic accuracy for NAFLD advanced fibrosis in higher prevalence tertiary center cohorts. In community-based and other low prevalence cohorts, blood-based NILDA are useful for excluding advanced fibrosis with high NPV but require additional non-invasive tests to improve their PPV.

TABLE 7 NAFLD fibrosis score for diagnosis of advanced fibrosis

Author, year (reference)	Number of patients (% F3-4)	AUROC F3-4	Sensitivity/specificity $\leq 1.455^a$	Sensitivity/specificity $> 0.676^b$	Number of indeterminates (%)	Comments and subgroups
Angulo, 2007 ^[43]	480 (26)	0.88	0.82/0.77	0.51/0.98	114 (24)	LR+ 11-26 (high cutoff)
	253 (29)	0.82	0.77/0.71	0.43/0.96	70 (28)	–LR 0.23–0.32 (low cutoff)
Qureshi, 2008 ^[65]	331 (14)	N/A	0.96/N/A	N/A/0.84	154 (46)	
Wong, 2008 ^[66]	162 (11)	0.64	0.39/0.81	0/0.99	32 (20)	
Wong, 2010 ^[67]	228 (23)	0.75	0.73/0.69	0.18/0.96	N/A	
McPherson, 2010 ^[68]	145 (19)	0.81	0.78/0.58	0.33/0.98	N/A	
Ruffillo, 2011 ^[69]	138 (27)	0.68	0.23/N/A	N/A/1.0	42 (30)	
Xun, 2012 ^[70]	154 (16)	0.65	0.37/0.86	0.08/1.0	25 (16)	
Sumida, 2012 ^[71]	576 (11)	0.86	0.92/0.63	0.33/0.96	206 (36)	
Cichoz-Lach, 2012 ^[72]	126 (21)	0.92	0.96/N/A	N/A/0.84	39 (31)	
Yoneda, 2013 ^[73]	235 (16)	0.84	N/A	0.68/0.88	N/A	Normal ALT cohort
Lee, 2013 ^[74]	107 (32)	0.88	0.82/0.77	N/A	N/A	
Demir, 2013 ^[75]	Aqsw'daZ	0.96	0.75/0.93	0.19/1.0	16 (13)	
Cui, 2015 ^[76]	102 (19)	0.82	0.84/0.69	0.21/0.96	N/A	
Lykiardopoulos, 2016 ^[77]	158 (24)	0.79	0.44/N/A	N/A/0.37	84 (53)	
Rath, 2016 ^[78]	60 (3)	0.47	0.05/N/A	N/A/1.0	8 (13)	
Jun, 2017 ^[79]	328 (18)	0.64	0.53/0.67	0.09/0.98	N/A	
McPherson, 2017 ^[80]						Age (y) ≤ 35
	74 (11)	0.52	0/0.91	0/1.0	N/A	36–45
	96 (19)	0.86	0.78/0.80	0.22/1.0		46–55
	197 (22)	0.81	0.81/0.65	0.22/0.97		56–64
	191 (34)	0.83	0.95/0.44	0.31/1.0		≥ 65
	76 (40)	0.81	0.93/0.20	0.57/0.85		
Bertot, 2018 ^[81]	241 (31)	0.72	N/A	0.76/0.85	N/A	
Patel, 2018 ^[82]						Age (y)
	115 (10)	0.72	0.09/0.35	0.45/0.98	N/A	< 50 y
	154 (34)	0.76	0.02/0.62	0.68/0.83		50–64
	60 (46)	0.71	0.04/0.84	0.74/0.68		≥ 65
Chan, 2019 ^[83]	753 (24)	0.69	N/A	0.16/0.99	215 (29)	
Kaya, 2019 ^[84]	463 (17)	0.71	0.71/0.63	0.15/0.96	173 (37)	
Yang, 2019 ^[85]	453 (28)	0.53	N/A	0.19/0.92	N/A	
Anstee, 2019 ^[86]	2417 (80)	0.74	0.89/0.37	0.38/0.89	1208 (51)	Clinical trial cohort
Petta, 2019 ^[54]	968 (28)	0.76	0.74/0.70	0.16/0.97	348 (36)	
De Carli, 2020 ^[87]	246 (9)	N/A	N/A	0.12/0.96	N/A	Bariatric surgery cohort
Bril, 2020 ^[88]	213 (17)	0.64	N/A	0.91/0.40	144 (68)	
Alkayyali, 2020 ^[89]	166 (29)	0.73	0.75/0.47	0.25/0.93	79 (47)	DM
	183 (10)	0.72	0.85/0.60	0/0.97	77 (42)	Non-DM
						Age (y)
Pitisuttithum, 2020 ^[90]	472 (6)	0.68	0.67/0.65	0.10/0.94	N/A	< 60
	131 (17)	0.65	0.74/0.41	0.26/0.86	N/A	≥ 60

^aLower cutoff to rule-out F3-4.^bhigher cutoff to rule-in F3-4.

Abbreviations: ALT, alanine aminotransferase; AUROC, area under receiver operating characteristic curve; DM, diabetes mellitus; LR, likelihood ratio; N/A, not available/not applicable.

ALD

Assessment of the diagnostic utility of blood-based NILDA in ALD is limited due to small study cohorts with variable severity of alcoholic hepatitis, biopsy sampling, and histologic scoring systems. A study in 218 patients with ALD indicated that indirect markers such as APRI have low diagnostic accuracy for F2-4 or cirrhosis (AUROC 0.59–0.67), but proprietary tests such as FibroTest™, FibroMeter™, or HepaScore™ had better performance for detection of F2-4 (AUROC 0.83) and cirrhosis (AUROC 0.92–0.94).^[107] A systematic review that included eight studies with blood-based marker panel assessment of advanced fibrosis or cirrhosis in patients with ALD also reported high accuracy for FibroTest, FibroMeter™, HepaScore™, and ELF™ for cirrhosis, but significant heterogeneity among studies precluded summary analysis.^[108] Based on our systematic review,^[94] there were too few studies to allow for recommendation regarding use of blood-based NILDA for ALD.

Other CLD

Similar to HCV mono-infection, NILDA tests are also important for the determination of liver disease severity in patients with HIV-HCV co-infection prior to DAA therapy. Our systematic review identified 12 studies, mostly reporting results for APRI and FIB-4.^[94] In general, blood-based markers appear to have similar diagnostic performance for significant fibrosis to patients who were HCV mono-infected, with fewer studies identified for the detection of advanced fibrosis and cirrhosis.

Post-SVR diagnostic limitations for blood-based NILDA also apply to HIV-HCV co-infection. Reduced blood-based NILDA accuracy due to associated thrombocytopenia, or potential antiretroviral therapy-related changes in bilirubin and GGT, need to be considered while interpreting these tests.^[109]

Few studies have assessed the diagnostic role of blood-based biomarkers for staging fibrosis in chronic cholestatic diseases and have included mostly patients with PBC.^[110] APRI and FIB-4 are the most frequently used simple nonproprietary tests. A study of 103 patients with PBC indicated AUROC of 0.77 to 0.93 for $\geq$ F2 for APRI and FIB-4, with better performance for the detection of cirrhosis.^[111] However, disease-specific diagnostic thresholds have not been established for blood-based tests.^[111–113] In a study of 229 patients with PSC, ELF and FibroTest had AUROC > 0.8 for the detection of F4 but were comparable with simple tests.^[114] In general, blood-based markers have acceptable accuracy for diagnosing cirrhosis related to chronic cholestatic disease;

however, the clinical utility of blood-based NILDA tests for staging fibrosis, especially in less advanced stages of fibrosis, in these patients is less certain than for viral hepatitis or NAFLD.

PICO 2: In adult patients with chronic liver disease, including hepatocellular (HCV, HIV-HCV, HBV, HIV-HBV, NAFLD, and ALD) or cholestatic (PSC and PBC) disorders, is any blood-based biomarker panel superior to another blood-based biomarker panel in staging hepatic fibrosis (F0-1 vs. F2-4, F0-2 vs. F3-4 and F0-3 vs. F4) using histopathology as the reference?

Guideline Statements

4. In patients with chronic HCV who require fibrosis staging, AASLD recommends using simple, less costly, and readily available blood-based NILDA such as FIB-4 over complex proprietary tests (strong recommendation, moderate quality of evidence).
5. In patients with NAFLD who require fibrosis staging, AASLD recommends the use of simple, less costly, and readily available blood-based NILDA tests such as FIB-4 or NAFLD fibrosis score over complex proprietary tests for the detection of advanced fibrosis (F3-4; strong recommendation, moderate quality of evidence).

TECHNICAL REMARKS

- Blood-based NILDA: Head-to-head studies comparing blood-based NILDA in the same patient population are limited in number. In comparing one study to another, the pooling of sensitivity and specificity may be suboptimal because different thresholds have been used across typically heterogeneous populations and settings. Other assessments (e.g., predictive values) depend on the clinical setting and prevalence of different fibrosis stages in the population being studied. Most of the research studies were developed in patient populations from tertiary or referral centers, which limits generalizability.
- In chronic HBV prior to therapy, there are limited data comparing simple with proprietary NILDA.
- There are limited data in diseases other than viral hepatitis and NAFLD that directly compare blood-based NILDA.

BACKGROUND

Blood-based NILDA have been studied predominantly in patients with HCV and NAFLD. In addition, comparison is usually only between select blood-based markers and involves a variety of cutoffs. This makes recommending one marker over the other difficult, especially for intermediate stages. In general, all blood-based markers are more accurate at identifying the absence of fibrosis or the presence of cirrhosis than intermediate stages of fibrosis. The diagnostic performance of proprietary and nonproprietary tests is not significantly different in clinical practice. Although proprietary markers may be suitable in select situations, nonproprietary tests are readily available, repeatable, and less expensive than proprietary tests.

Several studies have compared APRI with an alternate blood-based NILDA with a paired liver biopsy across liver disease diagnoses.^[94] The performance of proprietary and nonproprietary tests compared with APRI was not significantly different for F0-1 versus F2-4, F0-2 versus F3-4, and F0-3 versus F4 across select cutoffs. However, limitations include the following: 1) lack of comparison across all cutoffs; 2) few studies that do not have APRI as a comparator group; and 3) limited studies for proprietary markers in comparison to each other.

EVIDENCE AND RATIONALE

HCV

Studies have examined the role of blood-based NILDA predominantly in the pre-DAA era. Overall, proprietary and nonproprietary blood markers have comparable diagnostic accuracies for significant fibrosis.^[115] Comparative data are largely limited to APRI, FIB-4, and FibroTestTM because these markers have the most complete data. Less comparative data are available for ELFTM, FibrometerTM, Fibrospect IITM, and Kings; however, sensitivities and specificities of these tests are not significantly different compared with the aforementioned tests. For the presence of significant fibrosis, the DOR range is from 5.44 to 13.35 and not significantly different among APRI (cutoff 0.5 or 1), FIB-4 (cutoff 1.45), Fibrometer (cutoff 0.5), and FibroTest (cutoff 0.48). APRI (cutoff 1) had the highest DOR 13.35 (6.7–26.57). For presence of advanced fibrosis, the DOR range is 6.87 to 21.49, with similar performance for APRI (cutoff 1.5), FIB-4 (cutoff 3.25), and FibroTest (0.48), as well as FIB-4 (cutoff 1.45 or 3.25) and ELF (cutoff 9.13–9.49). ELF had the highest DOR (21.49 [8.43–54.75]) [94]. In a large observational cohort (>2000 paired biopsy measurements), FIB-4 (0.83 [95% CI: 0.81–0.85]) and APRI (0.80 [95% CI: 0.78–0.82]) had equivalent performance.^[116] In

another study, FIB-4 correctly classified a higher proportion of patients even though the overall performance of APRI and FIB-4 was similar.^[117] Single-center studies have suggested that there may be overestimation in fibrosis in African American individuals using FibroSpect II, FIB-4, and APRI^[118] and inaccurate results in patients with normal transaminases, especially in the presence of end stage renal disease.^[58,119]

HBV

APRI and FIB-4 have the most complete data available, although proprietary markers (e.g., FibroTestTM) may also have similar performance in predicting cirrhosis.^[50,120–122] For the presence of advanced fibrosis, the DOR ranged from 4.86 to 9.28 and was not significantly different for APRI (cutoff 0.5) and FIB-4 (cutoff 1.45). FIB-4 (cutoff 2.2) had the highest DOR. However, there are concerns that APRI and FIB-4 cutoffs may not be applicable across all populations, and there may be a high risk of misclassification, especially with current cutoffs.^[122–125]

NAFLD

There are limited data comparing the DOR across the various tests. FIB-4 (using cutoff 1.45 to rule out or 2.67 to rule in) had a higher DOR than APRI (using cutoff 1.5), but data were not available to compare DOR for other tests.^[94] There was insufficient data to compare DOR for other tests such as FibroTest (cutoff 0.70) or ELF (cutoff 9.8).

Nonproprietary tests such as FIB-4, APRI, and NFS help to rule-out advanced fibrosis.^[125] Nonproprietary tests scores have generally similar performance in excluding advanced fibrosis, although, in select studies, NFS and FIB-4 may have better performance characteristics.^[68,103,126] Cutoffs may need to be modified for select populations such as those who have class III obesity,^[127] and scores do not have adequate performance characteristics across all demographics.^[128–130] Performance also varied by age with increased sensitivity and decreased specificity of blood-based markers with age.^[80,86] There are conflicting data on the diagnostic accuracy of proprietary fibrosis panels (e.g., Fibrometer and ELF) compared with FIB-4 and NFS for the detection of fibrosis in NAFLD.^[106,131,132]

Other CLD

In patients with HCV/HIV co-infection, the sensitivities and specificities of APRI, FIB-4, and FibroTest were

not significantly different for significant fibrosis, advanced fibrosis, and cirrhosis.^[94] The DOR was high for APRI for both significant fibrosis (DOR 3.9–5.5) as well as cirrhosis (DOR 15.24). Although smaller studies have shown that ELF and FibroTest performances were superior to nonproprietary tests (FIB-4 and APRI), there are not enough studies to recommend one test over the other.^[133,134] There are concerns that the performance of blood-based markers in individuals who are co-infected is not the same as compared with patients who are mono-infected with HCV.^[135]

Comparative data using blood-based NILDA for ALD, PBC, and PSC are limited. In a prospective study in patients with ALD, ELF (cutoff 10.5), and FibroTest (cutoff 0.58) identified advanced liver fibrosis in both primary and specialty care with high diagnostic accuracy and outperformed nonproprietary markers (FIB-4 and APRI).^[136] However, all tests (proprietary and nonproprietary) had an AUROC > 0.8. Proprietary markers slightly overestimated the probability of advanced fibrosis in patients from primary care, showing that the studies of accuracy likely had selection bias toward patients with more advanced fibrosis. In small studies in patients with PBC, both nonproprietary (FIB-4 and APRI) and proprietary markers (FibroTest and ELF) may have been comparable in staging fibrosis.^[137,138] APRI and FIB-4 have been studied in other liver diseases such as hemochromatosis. For example, a recent study in 181 C282Y homozygotes for the hereditary hemochromatosis gene showed both APRI and FIB-4 to have excellent performance (AUROC 0.86–0.88) with 81% accuracy in predicting advanced fibrosis.^[139]

QUALITY OF EVIDENCE AND OTHER CONSIDERATIONS

A meta-analysis supporting PICO 2 provided imprecise diagnostic estimates and was derived from studies that mostly had a low risk of bias.^[94] The quality of evidence was judged to be moderate for sensitivity and specificity estimates.

PICO 3: In adult patients with chronic liver disease, including hepatocellular (HCV, HIV-HCV, HBV, HIV-HBV, NAFLD, and ALD) or cholestatic (PSC and PBC) disorders, is the combination of two blood-based biomarker panels superior to a single one for staging fibrosis (F0-1 vs. F2-4, F0-2 vs. F3-4, and F0-3 vs. F4) using histopathology as the reference?

Guidance Statements

6. In patients with chronic untreated HCV, AASLD suggests a sequential combination of blood-based markers may perform better than a single biomarker for F2-4 or F4 (ungraded statement).
7. In patients with NAFLD, AASLD suggests the sequential combination of blood-based NILDA may be considered for diagnosis of advanced fibrosis (F3-4) over using a single test alone (ungraded statement).

TECHNICAL REMARKS

- Very few studies are available that have solely compared the combination of serum biomarkers to a single biomarker in assessing fibrosis with histopathology as reference.
- Because simple single blood-based NILDA such as APRI, FIB-4, and NFS with upper and lower cutoffs frequently have indeterminate results, adding a second blood-based test may help to better classify patients according to fibrosis severity.
- Analyses supporting PICO 3 provided imprecise diagnostic estimates and were derived from studies that mostly either had a high or unclear risk of bias. The quality of evidence was judged to be low for sensitivity and specificity estimates
- For identifying patients with NAFLD advanced fibrosis, AASLD recommended a sequential approach with FIB-4 followed by imaging NILDA or ELF in FIB-4 ≥ 1.3 when available.^[3,4,140]

EVIDENCE AND RATIONALE

HCV

In an international multicenter study involving 2035 untreated patients and using sequential algorithms that combined APRI and FibroTestTM, the diagnostic accuracy was higher in detecting significant fibrosis F2-F4 (90%) and cirrhosis F4 (92%) compared with either test alone (65%–82%).^[141] In HCV, when combined, APRI and FIB-4 have excellent NPV to exclude advanced fibrosis.^[142]

HBV

Several studies have addressed various combinations of blood-based markers, but most of these have been performed in combination with imaging-based elastography. In one study, the combination of FIB-4 and APRI had limited sensitivity (<64%) for F2-4 or F3-4.^[143] A combination of five blood-based markers achieved an acceptable diagnostic accuracy of 76% in a small sample size of 70 patients with HBV. Sensitivity, specificity, PPV, and NPV were 87%, 70%, 60%, and 91%, respectively, for significant fibrosis.^[144]

NAFLD

In a study using sequential analysis, the combination of FIB-4 and ELF did not achieve better diagnostic accuracy than FIB-4 alone.^[131] Using various cutoffs, a meta-analysis showed that a combination of NFS and FIB-4 is better than BARD (a score derived from the BMI, AST/ALT ratio, and presence of type 2 diabetes mellitus [T2DM]) alone.^[126] Another study in 407 patients with NAFLD indicated that the parallel combination of NFS + FIB-4 resulted in an AUC of 0.81 for F3-4 but with higher misclassification/indeterminate rate of 54%.^[105] The sequential combination of FIB-4 and NFS resulted in a lower AUC of 0.77 but reduced misclassification/indeterminate rates to 28%.^[126] Data from large NAFLD clinical trial cohorts have indicated that the simultaneous use of two noninvasive tests such as NFS or FIB-4 and ELF result in high sensitivity and specificity (0.89–0.99) but were associated with an increased proportion of patients (66%–92%) with nondiagnostic or indeterminate results.^[86,128] There are conflicting data on the diagnostic accuracy of proprietary fibrosis panels (e.g., Fibrometer and ELF) compared with FIB-4 and NFS for detection of fibrosis in NAFLD.^[106,129,130] In a prospective study of patients with NAFLD in primary care, sequential testing using FIB-4 followed by ELF detected more advanced fibrosis/cirrhosis cases and reduced unnecessary referrals from primary care to secondary care by 80%. However, this pathway was only applicable to approximately one-half of the referrals. Sequential or two-tiered pathways also improved resource utilization.^[145,146] Novel NASH biomarkers, including markers of apoptosis and cell death, metabolomic and lipidomic markers, oxidative markers, and several combinations, are currently being studied; however, none as yet are sufficiently accurate to be used clinically.^[147]

Other CLD

For other chronic liver diseases such as ALD and PBC, no studies as of yet have addressed the question of

whether the combination of serum markers is better than a single biomarker with liver histology being the reference.

PICO 4: In adult patients with chronic liver disease, including hepatocellular (HCV, HIV-HCV, HBV, HIV-HBV, NAFLD, and ALD) or cholestatic (PSC and PBC) disorders, do serial blood-based biomarker panels accurately predict the natural history of progression of fibrosis or regression of fibrosis in response to therapy relative to serial histopathology as the reference?

Guidance Statement

8. AASLD suggests against the use of blood-based NILDA tests to follow progression, stability, or regression in histologic stage (as determined by biopsy) in chronic liver disease (ungraded statement).

TECHNICAL REMARKS

- There are a limited number of blood-based biomarker/longitudinal biopsy studies in HCV from the interferon (IFN) era. There are no studies to assess changes in blood-based biomarkers and fibrosis stage, as determined by biopsy, with DAA therapy. As a result, the optimal interval for repeat measurements for blood-based biomarkers post-SVR is not established.
- There are a small number of longitudinal biopsy studies in HIV-HCV cohorts with variability in the interval among biopsy assessments, scoring systems, and the types of anti-retroviral and HCV antiviral therapy.
- A limited number of studies have assessed biomarker changes with histology following antiviral therapy in patients with HBV. There are no studies that have assessed both serial biomarkers and paired biopsy histologic assessment in other chronic hepatitis cohorts (such as HBeAg positive [immunotolerant phase] or negative [inactive carrier phase] infection).
- Very few paired biopsy studies have been done to assess NILDA in other CLD.

BACKGROUND

Liver fibrosis can regress after therapy to reduce the precipitating factor (inflammation, necrosis, steatosis, and/or iron overload; [Table 8](#)).^[95,96,114,124,125,148–174]

The terms regression, reversion, and reversal are intended to indicate that fibrosis, even in the setting of histological cirrhosis, decreases. However, these terms are not intended to indicate that the liver returns to normal in architecture and/or fibrosis content, especially in the setting of histologic cirrhosis.^[38,173] Most of the evidence demonstrating fibrosis regression and/or cirrhosis comes from studies that have analyzed large cohorts of patients with HBV or HCV following antiviral therapy.^[174–179] There is increasing evidence for the reversibility of fibrosis in NAFLD, but there remains a relative paucity of longitudinal histologic data with blood-based biomarkers for other liver diseases. One of the major limitations of currently available blood-based biomarkers is that they often misclassify patients with intermediate stages of fibrosis^[52,180] and are not able to differentiate adjacent stage disease.^[181] Importantly, extracellular matrix deposition and degradation is not a linear process and varies based on disease etiology.^[182,183] These factors limit the ability of blood-based biomarkers to follow the progression or regression of fibrosis across the spectrum of liver disease.

EVIDENCE AND RATIONALE

HCV

In the DAA era, there has been greater dependence on noninvasive tests, both pre- and post-treatment, to assess liver fibrosis stage. Blood-based biomarker scores appear to decline during treatment and immediately following SVR,^[184–187] suggesting that biochemical responses may influence these indices during and immediately following antiviral therapy. Thus, routine use of blood-based biomarkers based on liver inflammation after SVR in patients with advanced fibrosis or cirrhosis is likely to be associated with a substantial underestimation for significant fibrosis,^[95,96] and there are no validated data on the degree of improvement in post-SVR biomarker thresholds that correlate with fibrosis regression.^[28]

Although prior studies have assessed both histology and blood-based biomarkers following antiviral therapy in HCV, biomarker associations with fibrosis progression or regression are largely derived in the setting of IFN-based therapy^[148–150,153,156,161] or from maintenance IFN and other antifibrotic therapy in virologic nonresponders.^[151,152,154,155] We could not identify large studies with long-term follow-up in patients receiving DAA therapy that included paired biopsy and

biomarkers. Paired biopsy and biomarker studies in patients coinfecting with HIV-HCV have included mixed cohorts with HCV mono-infection, various IFN-treatment regimens, and variable intervals of histological assessment.^[157–162] Only a few studies have reported changes in biomarker indices with fibrosis stage. APRI, FIB-4, or FibroTest algorithms are the most frequently assessed biomarkers ([Table 5](#)). The fibrillary collagen formation marker procollagen type III (Pro-CIII) was associated with histologic fibrosis progression at 52 weeks in a chronic HCV nonresponder cohort receiving antifibrotic therapy, but this finding requires validation in other HCV paired-biopsy cohorts.^[154] A recent study utilizing both baseline and follow-up FIB-4 after SVR with DAA along with baseline albumin and GGT had acceptable performance (time-dependent AUROC of 0.72–0.74) in excluding those who develop HCC within 3 years,^[188] suggesting that blood-based NILDA may be used in the future to help risk-stratify patients for HCC surveillance after SVR.^[188–194]

HBV

Antiviral therapy in HBV results in viral suppression and fibrosis regression, including reversal of cirrhosis.^[175,179] Despite the low cost, ease of interpretation, and access advantages in resource-limited settings, simple markers such as APRI and FIB-4 are not able to follow changes in fibrosis. In a cohort of 294 patients receiving antiviral therapy with paired-biopsy assessment, APRI and FIB-4 did not correlate with histologic fibrosis regression observed at 5 years.^[123] Biomarkers incorporating transaminases or acute phase reactants will likely demonstrate early biochemical responses that may not reflect histologic regression following antiviral therapy in HBV, resulting in false-negative tests.

NAFLD

The current regulatory landscape requiring assessment of histologic efficacy endpoints in NAFLD therapeutic development has resulted in an increasing number of paired biopsy and biomarker studies reported from large clinical trials ([Table 8](#)). The most frequently assessed biomarkers include NFS, FIB-4, APRI, and ELF™. Longitudinal data from the NASH CRN on 292 patients with paired biopsies over a median of 2.6 years indicated modest AUROCs (0.66–0.73) for predicting fibrosis progression using simple markers such as FIB-4, APRI, and NFS; fibrosis scores adjusted for baseline fibrosis stage were associated with progression, but not regression, of fibrosis.^[125] The prevalence of significant fibrosis was 50% in this study, and the utility of these simple markers alone or in combination with other

TABLE 8 Serum biomarkers for fibrosis progression and regression

Serum biomarker, year of study (reference)	Etiology and baseline fibrosis prevalence	Paired biopsy (n)	Sampling interval	Fibrosis change from baseline	Change in index biomarker scores with change in fibrosis stage	Comments
PIIINP and HA, 2001 ^[148]	HCV (F2-4 = 38% for n = 105 NR)	239	16–26 mo	No significant change in Knodell/METAVIR stage	No change in fibrosis or serum markers	Data based on response to IFN-based therapy
FibroTest™, 2002 ^[149]	HCV (F3 = 32%, F4 = 0%)	134	72 wk	Progression (n = 28) No change (n = 83) Regression (n = 23)	Progression: 0.04 No Change: -0.02 Regression: -0.03	IFN-based therapy; Knodell score (no stage F2)
FibroTest™, 2003 ^[150]	HCV (F2 = 17%, F3 = 6%, F4 = 6%)	352	72 wk	Progression (n = 61) No change (n = 193) Regression (n = 98)	Progression: +1 stage = -0.06, +2 = 0.02, +3 = -0.01 No change: -0.07 Regression: -1 stage = -0.09, -2 = -0.15, -3 = -0.25	IFN-based therapy; N = 32 F4; FT decline significant in 17/32 ≥ 1 stage decrease. No change in FT for n = 15/32 with F4 at follow-up
HA, TIMP-1, PIIINP, YKL-40, 2010 ^[151]	HCV (Ishak 4 = 30%)	209	24–48 mo	Progression n = 70 (34%)	Not provided	HALT-C IFN-based therapy. Baseline HA and platelets significant in multivariate model for fibrosis progression
FibroTest™, 2013 ^[152]	HCV (F2 = 46%, F3 = 54%)	258	3.6–3.9 y	Progression (n = 97) No change (n = 111) Regression (50)	Progression: +1 stage = 0.04, +2 = 0.07, +3 = 0.23 No change = 0.03 Regression: -1 stage = 0.01, -2 = 0.01, -3 = -0.01	EPIC-3 IFN-based therapy. No association between FibroTest and differences in fibrosis stage
FibroSURE®, 2014 ^[153]	HCV (F2-4 = 48%)	133	72 wk	No change n = 80 (60%)	Change in FT/FS was not associated with change in fibrosis stage	IFN-based therapy
FibroTest™, 2014 ^[154]	HCV (Ishak 2 = 40%, 3 = 45%, 4 = 15%)	194	52 wk	Progression n = 34 (18%)	Not provided	HCV non-IFN Antifibrotic study; Pro-CIII associated with fibrosis progression in multivariate model
FIB-4, APRI, Forns Index, 2015 ^[95]	HCV (F0-1 = 60%, F2 = 27%, F3-4 = 13%)	115	5.9 ± 1.8 y	Progression (n = 5) No change (n = 106) Regression (n = 4)	Lower index scores for all markers at post-SVR biopsy	All patients with SVR. Optimal lower cutoffs associated with accuracy 71%-79% for F2-4, and 70%-83% for F3-4
FibroTest™, 2016 ^[155]	HCV (Ishak 2 = 39%, 3 = 44%, 4 = 15%, 5 = 1%)	201	52 wk	Progression (n = 42) No change (n = 122) Regression (n = 31)	Progression: +1 stage = -0.04, +2 = 0.00 No change = -0.03 Regression: -1 stage = 0.02	HCV in non-IFN antifibrotic study. No association with FibroTest index and changes in fibrosis stage
FIB-4, APRI, King score, ELF®, 2016 ^[96]	HCV (F4 = 100%)	38	61 (48–104) months	Regression (n = 23) No change (n = 15)	Lower index scores for all markers at post-SVR biopsy. AUROC for post-SVR F4	All patients with SVR. No difference in scores between regressors and non-

					APRI = 0.58, FIB-4 = 0.59, King score = 0.59, ELF = 0.63	regressors at post-SVR biopsy (AUROC 0.52-0.75)
ELF®, 2017 ^[156]	HCV (Ishak 3 = 14%, 4 = 14%, 5/6 = 26%)	70	24 mo	Progression (<i>n</i> = 21) No change (<i>n</i> = 25) Regression (<i>n</i> = 24)	ELF at baseline/12 months to predict 1-stage progression (AUROC 0.72) and regression (0.64)	IFN-based therapy
FibroSURE®, APRI, 2006 ^[157]	Mixed (HCV and HIV-HCV) (F2 = 28%, F3 = 7%, F4 = 3%)	119	4.2 (2.8–6) years	Progression <i>n</i> = 25 (21%)	FibroSure PPV 0.31 and APRI 0.375 for predicting F2-4 on second biopsy	IDU cohort; HIV-HCV = 27%
APRI, 2007 ^[158]	HIV-HCV (Ishak 3 = 11%, 4 = 1%)	174	2.9 y	Progression in <i>n</i> = 41 (24%)	AST but not APRI associated with fibrosis progression	
FibroTest™ Forns Index, APRI, FIB-4, HepaScore™, FibroMeter™, 2009 ^[159]	HIV-HCV (F2 = 46%, F3 = 23%, F4 = 11%)	114	72 wk	Progression (<i>n</i> = 37) No change (<i>n</i> = 49) Regression (<i>n</i> = 28)	Significant decline in all biomarker index scores with SVR, except HepaScore	Data based on IFN-based therapy response
FIB-4, APRI, 2010 ^[160]	HIV-HCV (Ishak 3 = 15%, 4 = 9%)	66	4.7 y	Progression (<i>n</i> = 21) No change (<i>n</i> = 26) Regression (19)	No difference in FIB-4 and APRI between progressors (Ishak ≥ 2) and no fibrosis change	
FibroMeter™, FibroTest™, HepaScore™, 2012 ^[161]	HCV and HIV-HCV (F3 = 25%, F4 = 27%)	101 (HCV <i>n</i> = 62, HIV-HCV <i>n</i> = 39)	96 wk	Progression (mean 0.2 METAVIR units)	Not provided	IFN-based therapy Progression in area of fibrosis, FibroMeter, and CirrhoMeter
FIB-4, APRI, 2014 ^[162]	HIV-HCV (F2 = 11%, F3 = 3%)	282	2.5 y	Progression <i>n</i> = 97 (34%)	Not provided	AST and ALT > 2.5 ULN between biopsies associated with fibrosis progression in multivariate model
FIB-4, APRI, FibroTest™, 2015 ^[163]	HIV-HCV (F0-F3)	38	3 y	Progression (<i>n</i> = 10) No change (<i>n</i> = 27) Regression (<i>n</i> = 1)	Progression: FIB-4 + 0.75, APRI + 0.36, FT + 0.04 No change/regressor: FIB-4: -0.06, APRI: -0.30, FT: -0.03	Only <i>N</i> = 5 with HCV treatment; differences between progressors and non-progressors for APRI and FIB-4 (<i>p</i> = 0.03); FT = not significant
FibroTest™, 2009 ^[164]	HBV (F2-4 = 44%)	462	48 wk	Regression (0.16-0.30 mean METAVIR units)	Not provided	Antiviral therapy/placebo treatment; FibroTest improved in virologic responders with F2-4, and placebo
APRI, FIB-4, 2016 ^[124]	HBV (Ishak 3 = 23%, 4 = 10%, 5-6 = 24%)	294	240 wk	Regression in F4-6 from 34% to 12%)	No correlation with regression	On antiviral therapy; 81%-89% baseline advanced fibrosis or cirrhosis missed by simple scores
APRI, FIB-4, 2019 ^[165]	HBV (median Ishak 3)	80	2.06 y to second biopsy	Regression 0.18 Ishak Units/year	Not provided	Multiple biopsies over 17 y, variable treatment, Greater relative decline FIB-4 (-17%) and APRI (-43%) in year 1

TABLE 8. (continued)

Serum biomarker, year of study (reference)	Etiology and baseline fibrosis prevalence	Paired biopsy (n)	Sampling interval	Fibrosis change from baseline	Change in index biomarker scores with change in fibrosis stage	Comments
APRI, FIB-4, NFS, BARD, 2010 ^[166]	NAFLD (F3-4 = 4%)	52	36 mo	Progression (n = 14) No change (n = 25) Regression (n = 13)	Progression: APRI = +0.003, FIB-4 = +0.079, NFS = +0.06, BARD = 0 No change/ Regression: APRI = -0.029, FIB-4 = -0.019, NFS = -0.017, BARD = 0	Prospective study; No significant correlation between change in fibrosis stage and markers
APRI, 2012 ^[167]	NAFLD (Any fibrosis = 45%)	78	Variable	Not provided Any fibrosis n = 22 (31%)	Baseline APRI = 0.29 After weight loss APR1 = 0.29	Bariatric surgery cohort with morbid obesity. Variable biopsy interval after weight loss. No change in APRI
APRI, FIB-4, NFS, 2017 ^[168]	NAFLD (F3-4 = 10%)	261	52 wk	Progression (n = 45) No change (n = 165) Regression (n = 51)	Progression: APRI = -0.16, FIB-4 = -0.05, NFS = +0.02 No change: APRI = -0.14, FIB-4 = -0.08, NFS = -0.42 Regression: APRI = -0.25, FIB-4 = -0.23, NFS = -1.00	Lifestyle intervention study
ELF™, FibroTest™, NFS, 2018 ^[169]	NAFLD (NASH CRN F3 = 46%, F4 = 54%)	427	96 wk	F3 : Progression (n = 41) Regression (n = 40) ; F4: Regression (n = 22)	No significant change in serum markers with fibrosis stage	Phase lib study
ELF™, FibroTest™/FibroSure®, 2018 ^[170]	NAFLD (F2 = 35%, F3 = 65%)	72	24 wk	Progression (n = 23) No change (n = 34) Regression (n = 23)	No change in serum markers across treatment groups	Phase II study for NAFLD stage F2-3
APRI, FIB-4, NFS, 2019 ^[124]	NAFLD (F3-4 = 26%)	292	2.6 y	Progression (n = 92) No change (n = 126) Regression (n = 74)	Progression: APRI = +0.2, FIB-4 = +0.5, NFS = +0.7 No change: APRI = -0.2, FIB-4 = +0.1, NFS = +0.4 Regression: APRI = -0.3, FIB-4 = 0.0, NFS = +0.5	NASH CRN cohort. APRI, FIB-4, and NFS associated with progression, but not regression
ELF™, 2020 ^[171]	NAFLD (F3 = 44%, F4 = 4%)	43	12 wk	Regression (n = 14)	Decline in ELF (-7% vs. -3%) and Pro-CIII (-56% vs. -9%) for histologic responders vs. non-responders	Phase II study

FIB-4, APRI, FibroSURE®, ELF™ 2019, 2022 ^[172,173]	NAFLD (F3 = 56%)	931	18 mo	Progression (n = 130) No change (n = 412) Regression (n = 223)	AUROC 0.58-0.61 for 10% decrease in markers at month 18 to predict fibrosis regression	Phase III study Data provided by treatment groups indicate greater decline in markers with regression. Overall weak association between improvement in markers and fibrosis stage
ELF™, FibroTest™, 2019 ^[174]	NAFLD (F3 = 52%, F4 = 47%)	1527	48 wk	Regression (n = 207) No histologic response (n = 1324)	Response (regression): ELF = -0.2%; FT not provided No response: ELF = 1.3%; FT not provided	Pooled Phase III data. Data provided as fibrosis regression and no worsening NASH (histologic response)
ELF™, FibroTest™/FibroSURE®, 2019 ^[114]	PSC (Ishak 4-6 = 26%)	234	96 wk	Progression (n = 80) No change (n = 74) Regression (n = 79)	Not provided	Phase II study. Baseline ELF associated with progression to cirrhosis

Abbreviations: APRI, AST-to-platelet Ratio Index; AUROC, Area Under Receiver Operating Characteristic Curve; BARD, body mass index, AST/ALT ratio, and presence of type 2 diabetes mellitus; ELF, enhanced liver fibrosis; FT/FS, FibroTest/FibroSURE; HA, hyaluronic acid; HAL T-C, hepatitis C antiviral long-term treatment against cirrhosis; HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus; IFN, interferon; NAFLD, nonalcoholic fatty liver disease; NASH, nonalcoholic steatohepatitis; NFS, NAFLD fibrosis score; PBC, primary biliary cholangitis; PLINP, amino-terminal propeptide of type III procollagen; Pro-C3, N-terminal pro-peptide of type III procollagen; PSC, primary sclerosing cholangitis; TIMP-1, tissue inhibitor matrix metalloproteinase 1; ULN, upper limit of normal.

noninvasive tests, to follow fibrosis progression in lower prevalence settings, remains to be determined. A phase IIb study for NASH CRN stage 3 and 4 noted an improvement in histologic fibrosis by one stage in 18% to 23% of stage 3 patients and in 8% to 13% of patients with baseline cirrhosis.^[195] Progression to cirrhosis was observed in 19% to 22% at 96 weeks across the treatment groups. Despite these histologic changes, there were no significant differences observed between the treatment and placebo groups through week 96 in liver biochemistry, ELF score, FibroTest, or NFS.^[169] A 12-week clinical trial in 43 patients with NAFLD (including 48% with advanced fibrosis) reported significant reductions in PRO-C3 and ELF in patients with histologic response (including improvement in NASH) compared with nonresponders, but a corresponding change in scores with change in fibrosis was not provided.^[196] In an ongoing phase III study of 931 patients with NAFLD with stage F2 or F3, an interim analysis of biopsy and several blood markers (FIB-4, APRI, FibroTest, ELF, PRO-C3) indicated weak associations between change in markers and improvement in fibrosis stage at 18 months.^[140] Although multiple studies have noted improvement in NAFLD fibrosis stage following bariatric surgery for patients with class III obesity,^[197] very few have incorporated blood-based biomarkers to evaluate for associations with histologic resolution. As with other CLDs, biomarkers that incorporate liver transaminases and acute phase reactants (Table 5) will need to be interpreted with caution following therapies that may improve necroinflammation, but not fibrosis, over a relatively short study duration.^[198]

Other CLD

Although small studies in ALD and cholestatic disease have examined blood-based NILDA in cross-sectional assessments, for following disease progression or for determining prognosis, none have specifically evaluated blood-based biomarkers for following changes in fibrosis on biopsy. A recent phase II study in 234 patients with PSC evaluated FibroTest and ELF in relation to serial biopsy assessment at 96 weeks. Association and directional change in biomarker indices with observed fibrosis change at week 96 were not provided.^[114]

PICO 5: In patients with NAFLD, are blood-based biomarker panels accurate in grading hepatic steatosis (S0 vs. S1-3, S0-1 vs. S2-3, and S0-2 vs. S3) using histopathology or magnetic resonance (MR) spectroscopy (MRS) or magnetic resonance imaging (MRI)-proton density fat fraction (PDFF) as the reference?

Guidance Statement

9. AASLD suggests against the use of blood-based NILDA to detect steatosis in patients with NAFLD (ungraded statement).

TECHNICAL REMARKS

- In adult patients with CLD, Controlled attenuated parameter (CAP) and MRI can reliably quantify the degree of steatosis. MRI-PDFF and MRS have excellent correlation with histology for detecting and grading steatosis and can be used as reference standards.^[3]
- Steatosis, independent of fibrosis, is associated with increased systemic inflammation and has prognostic importance as a predictor of cardiovascular disease, DM, and, in severe cases, liver-related mortality.
- Patients with chronic liver disease associated with steatosis other than NASH, such as chronic HCV genotype 3, have not been well-studied.
- The available evidence is insufficient to make a recommendation as to which noninvasive test(s) or algorithm(s) should be used, compared with others, to assess steatosis.
- There is insufficient evidence to recommend blood tests as clinical endpoints to monitor changes in steatosis, independent of fibrosis over time.
- There is insufficient evidence to make a recommendation regarding a specific blood-based test or algorithm to use in combination with imaging-based testing for the assessment of steatosis.
- Because BMI is included in many of the indices, caution is necessary when using NILDA to assess steatosis in patients who have undergone bariatric surgery.

BACKGROUND

Although liver fibrosis assessment has been the focus of noninvasive tests in liver diseases, steatosis is also important in the assessment of disease severity in NAFLD. Histologically, steatosis (S) is graded 0 to 3 based on the proportion of hepatocytes that contain fat as follows: S0 (<5%), S1 (5%–33%), S2 (34%–66%), and S3 (>66%) steatosis (Table 3).^[21,22] In addition

to liver-related outcomes in NASH (decompensation, HCC),^[198,199] steatosis is associated with systemic inflammatory markers,^[200,201] DM,^[202–204] the metabolic syndrome,^[205] cardiovascular disease,^[203,204,206–209] and atherosclerosis.^[210] Several noninvasive algorithms have been developed to assess steatosis using biochemical and clinical variables.^[211,212] Although many steatosis algorithms have been developed or validated based on ultrasound (US)^[202,213–219] several have utilized histologic^[182,217–221] or MR-based assessments^[205,222,223] as the reference standard (Table 9). However, there are limited data to support longitudinal assessments of steatosis using these algorithms.^[25]

EVIDENCE AND RATIONALE

Most algorithms include standard liver-related blood tests (AST, ALT, bilirubin, GGT), blood tests associated with hyperlipidemia (triglycerides [TG], cholesterol), and conditions associated with steatosis (DM, increased BMI, increased waist circumference [WC], and the metabolic syndrome) in some combination (Table 9). Of note, some algorithms differ by sex. Table 10 summarizes the performance and cutoffs for algorithms to assess steatosis.^[202,205,217–223,225–227,229–232]

Fatty liver index (FLI)

This algorithm utilizes TG, BMI, WC, and GGT. Although initially developed in comparison to conventional B-mode US,^[214,217] FLI has also been validated against liver histology and MRI.^[205,218,219,221,230,233] Depending on the cutoff, studies have shown sensitivity ranges from 44% to 100%, whereas specificity ranged from 3% to 91% with AUROC 0.59 to 0.86. Furthermore, a FLI modified for North American patients (compared with non-North American patients) and including age, race and ethnicity, fasting insulin, and glucose seemed to perform better in a US population.^[234]

Hepatic steatosis index (HSI)

This algorithm includes AST, ALT, BMI, and GGT. Although initially developed in a cohort compared with US,^[213] HSI has also been validated against liver histology and MRI.^[204,218,221,230] Depending on the cutoff, HSI had a sensitivity ranging from 7% to 88%, specificity ranging from 9% to 93%, and AUROC 0.49 to 0.81. One advantage of HSI is its simplicity because it uses routine tests and does not require additional factors such as WC or insulin resistance to be measured. However, one limitation is that those with increased BMI, especially if over age

TABLE 9 Noninvasive algorithms to assess hepatic steatosis compared with histology or MR spectroscopy or MR PDFF

Algorithm	Formula or Components
FLI	$\text{Log}(0.953 \times \ln \text{ TG}) + 0.139 \times \text{BMI} + 0.718 + \ln(\text{GGT}) + 0.053 \times \text{WC} - 15.745 \times 100$
HSI	$8 \times \text{ALT}/\text{AST} + \text{BMI} + 2 \text{ (if DM)} + 2 \text{ (if female)}$
LAP	$(\text{WC [cm]} - 65) \times \text{TG (mmol/L)}$ male individuals $(\text{WC [cm]} - 58) \times \text{TG (mmol/L)}$ female individuals
NLFS	$-2.89 + 1.18 \times \text{MS} + 0.45 \times \text{DM} + 0.15 \times \text{insulin} + 0.04 \times \text{AST} - 0.94 \times \text{AST}/\text{ALT}$
ION	$1.33 \times \text{waist-to-hip ratio} + 0.03 \text{ TG (mg/dL)} + 0.18 \text{ ALT (U/L)} + 8.53 \text{ HOMA-IR} - 13.93$ in male individuals $0.02 \text{ TG (mg/dL)} + 0.24 \text{ ALT (U/L)} + 9.61 \text{ HOMA-IR} - 13.99$ in female individuals
Steatotest™	ALT, A2M, ApoA1, haptoglobin, total bilirubin, GGT, total cholesterol, TG, glucose, age, gender, BMI
TyG	$\text{Log}(\text{TG [mg/dL]}) \times \text{glucose (MG/dL)}/2$
VAI	$(\text{WC}/39.68 + 1.88 \text{ BMI}) \times (\text{TG}/1.03 \times 1.31/\text{HDL})$ for male individuals $(\text{WC}/36.58 + 1.89 \text{ BMI}) \times (\text{TG}/0.81 \times 1.52/\text{HDL})$ for female individuals
DSI	ALT, BMI, age, sex, triglyceride and glucose levels, diabetes, hypertension, and ethnicity

Abbreviations: A2M, α -2 macroglobulin; ALT, alanine aminotransferase; ApoA1, apolipoprotein A; AST, aspartate aminotransferase; BMI, body mass index; DM, diabetes mellitus; DSI, Dallas steatosis index; FLI, fatty liver index; GGT, gamma-glutamyl transferase; HDL, high-density lipoprotein; HOMA-IR, Homeostasis Model of Assessment For Insulin Resistance; HSI, hepatic steatosis index; ION, index of NALFD; LAP, lipid accumulation product; MS, metabolic syndrome; NAFLD, nonalcoholic fatty liver disease; NLFS, NAFLD liver fat score; PDFF, proton density fat fraction; TG, triglyceride; TyG, triglyceride index; VAI, visceral adiposity index; WC, waist circumference.

40 years, will have an increased HSI, which may explain its poor performance in some studies.^[221,231] Similar factors make HSI less reliable in the bariatric population.

Lipid accumulation product (LAP)

The lipid accumulation product was developed from the National Health and Nutrition Examination Survey to assess cardiovascular disease^[200] and has been used to detect hepatic steatosis.^[215] The index includes only two variables: WC and TG. The index has been compared with both liver biopsy^[220,230] and MR,^[216] with performance in assessing steatosis as a continuous variable with AUROC 0.68 to 0.73.

NAFLD liver fat score

The NAFLD liver fat score was developed against MRS and included the presence or absence of the metabolic syndrome and DM along with fasting insulin and AST and ALT.^[222] Depending on the cutoff,^[218,222,227] the sensitivity was 65% to 86%, specificity was 62% to 87%, and AUROC was 0.64 to 0.87.

Index of NAFLD

In a study of 152 patients with NAFLD from a cohort of 861 identified by increased echogenicity in the United States, the index of NAFLD (composed of waist-to-hip ratio, TG,

ALT, and Homeostatic Model Assessment of Insulin Resistance) was developed and compared with FLI.^[202] Depending on the cutoff, the sensitivity was 60% to 81%, specificity was 56% to 82%, and AUROC was 0.77.

SteatoTest®

This biomarker was developed based on the FibroTest™ and ActiTest® (AT), validated biomarkers for fibrosis and inflammation, respectively.^[182,220,235] SteatoTest includes the six components of FibroTest-AT (ALT, α -2 macroglobulin, apolipoprotein A-1, haptoglobin, total bilirubin, GGT) and adds BMI, total cholesterol, TG, and glucose adjusted for age and sex.^[214] This biomarker for steatosis has been used in those at high risk for NAFLD.^[217,225,227,236] One limitation of SteatoTest is the inclusion of total bilirubin, which can be increased in conditions such as Gilbert's syndrome. To overcome this, a modified version (SteatoTest-2®) has recently been developed that does not include BMI or bilirubin^[231] for those with increased unconjugated bilirubin or inaccurate or unavailable BMI. Depending on the cutoff, SteatoTest-2 has a sensitivity ranging from 38% to 90%, specificity ranging from 44% to 88%, and AUROC from 0.65 to 0.81.

TG-glucose index

The TG-glucose index was developed as a screening tool for insulin resistance.^[237] When used to determine whether NAFLD was present,^[218,228,229] it had an overall sensitivity of 70% to 94%, specificity of 60% to 92%, and AUROC of 0.68 to 0.90.

TABLE 10 Performance of blood-based algorithms for diagnosis of hepatic steatosis

Test	Reference	N	Cutoffs (if provided)	Comparator	Sensitivity	Specificity	AUROC
FLI	[224]	182	< 30	LB	100	3	0.59
			≥ 60		97	13	
	[201]	40	< 30 or ≥ 60	MR	90	74	0.86
	[214]	264	< 30 or ≥ 60	LB			0.75
	[216]	324	> 60	LB	76	87	0.83
	[221]	336	> 30	MR	75	69	0.79
			> 60		44	91	
	[217]	250	≥ 79	LB	81	49	0.67
	[199]	4458	< 30	LB	80		
HSI	[222]	135		LB		80	0.74
	[224]	182	< 30-45	LB	88	10	0.41
			≥ 36-67		7	90	
	[201]	40	< 30 or ≥ 36	MR	86	66	0.75
	[215]	364		LB			0.63
	[217]	324	> 41.6	LB	61	93	0.81
	[225]	366	35.6	LB	61	63	0.66
	[209]	10,724		LB	78	69	0.77
	[222]	135					0.71
LAP	[224]	182	Continuous	LB			0.63
	[215]	364		LB			0.70
	[221]	336		MR			0.78
NFLS	[220]	470	-0.640	MR	86	71	0.87
	[224]	182	-06.40	LB	71	62	0.64
	[226]	324	> 0.16	LB	65	87	0.80
ION	[199]	4458	< 11	LB	81	56	0.77
			≥ 22		60	82	
Steato-Test™	[220]	310	≥ 0.3	LB	90	54	0.79
			≥ 0.7		46	88	
	[225]	288	0.38	LB	86.9	50	0.65
			0.69		42	79	0.81
	[217]	494	0.38	LB	89	44	
			0.69		38	81	0.80
	[227]	220	0.52	MR	73	72	0.73
SteatoTest-2™	[225]	2997	0.40	LB	79	50	0.77
TyG	[218]	324	> 8.38	LB	80	92	0.90
	[228]	50	4.235	LB	94	69	0.86
	[229]	340	4.515	LB	70	60	0.68
VAI	[218]	324	> 1.25	LB	79	92	0.92

Abbreviations: AUROC, area under receiver operator characteristic curve; FLI, fatty liver index; HSI, hepatic steatosis index; ION, index of NAFLD; LAP, lipid accumulation product; LB, liver biopsy; MR, magnetic resonance; NAFLD, nonalcoholic fatty liver disease; NFLS, NAFLD liver fat score; TyG, triglyceride index; VAI, visceral adiposity index.

Visceral adiposity index

Increased visceral adiposity is associated with NAFLD.^[238–240] There are limited studies in NAFLD using liver histology as the reference standard.^[218] With a cutoff of 1.25, the visceral adiposity index showed a sensitivity of 79%, specificity of 92%, and AUROC of 0.92.

Dallas steatosis index

The Dallas steatosis index was developed from the Dallas Heart Study, a multiethnic, population-based, probability study of adults (age 18–65 y) to detect at least 5.5% steatosis by MRS.^[241] The index, which includes ALT, BMI, age, sex, TG

and glucose levels, DM, hypertension, and ethnicity, had a c-statistic of 0.824; it outperformed HSI (0.746) and overlapped with the FLI (0.810). However, the Dallas steatosis index has not been validated compared with liver histology as the reference standard.

PICO 6: In pediatric chronic liver disease (HCV, HBV, biliary atresia [BA], cystic fibrosis [CF] liver disease [CFLD], and NAFLD/NASH), are blood-based biomarkers accurate in staging hepatic fibrosis (F0-1 vs. F2-4, F0-2 vs. F3-4, and F0-3 vs. F4) using histopathology as the reference?

Guidance Statement

10. In the pediatric patients with chronic liver disease, AASLD suggests the use of simple, cost-effective, and readily available blood-based NILDA, such as APRI or FIB-4, for the detection of advanced fibrosis (F3-4) (ungraded statement).

TECHNICAL REMARKS

- Some blood-based NILDA in children have good accuracy in detecting advanced fibrosis but have difficulty discriminating earlier stages of fibrosis.
- FIB-4 does not perform as well in children as it does in adults, particularly very young children, due to the inclusion of age in the index.
- Rapid growth in children and attendant fluctuations in alkaline phosphatase can confound interpretation of blood or collagen-based NILDA tests in pediatric liver disease.
- There are insufficient biopsy validated data to recommend biomarkers for evaluating fibrosis in pediatric NASH and α 1AT at this time.
- In the pediatric population with CLD, there is growing but insufficient evidence to recommend blood-based NILDA as endpoints to monitor changes in fibrosis over time.

BACKGROUND

Inherited or acquired liver disorders of childhood such as BA, α 1AT, and CFLD often and uniquely progress

to cirrhosis and portal hypertension early in life. With the exception of NAFLD/NASH, HBV, and HCV, the majority of pediatric liver disorders that lead to advanced fibrosis and commonly require liver transplantation are hepatobiliary in nature. The rapid progression of liver disease in some children indicates a need to identify early markers of liver fibrosis to help facilitate early intervention. Markers empirically identified by genomic, proteomic, and metabolomic technologies, as well as targeted blood-based marker analysis, offer new strategies to predict outcomes in pediatric liver diseases. Putative growth-independent blood biomarkers reflecting matrix deposition, removal, and remodeling; hepatic stellate cell activation; collagen turnover; and chemoattractant expression in children with a variety of liver diseases have been identified.^[242–244]

Most blood biomarker studies in children, even when validated by liver biopsy, are single-center investigations. Furthermore, many direct blood-based biomarkers are confounded by rapid somatic growth in children with liver disease. Although evolving anti-fibrogenic therapies and novel markers/endpoints for clinical trials are being studied, there are currently limited data to support longitudinal assessments of fibrosis using blood biomarkers in children. APRI, FIB-4, and FibroTest™ have been the most commonly studied NILDA tests in children; there is much less information regarding other NILDA tests such as ELF™, Fibrometer™, Fibrospect II™, eLIFT, King's fibrosis score, and Hepascore as surrogates of liver fibrosis, as validated by histology in pediatric populations.

EVIDENCE AND RATIONALE

Each pediatric liver disorder has a distinct pathophysiology with both genetic and epigenetic origins. These disorders are clinically heterogeneous; therefore, the performance of blood biomarkers as surrogates of liver fibrosis must be studied and compared within individual disease groups rather than in conglomerate or even by biomarker.

BA

BA is a neonatal liver disease characterized by rapidly progressive fibro-obliteration of the biliary tract and is the leading indication for pediatric liver transplantation.^[245,246] In BA, fibrosis typically develops early in life and leads to cirrhosis before age 6 months (without Kasai portoenterostomy) and would be an ideal target for newly developed anti-fibrotic pharmacotherapies.^[246] The utility of APRI to assess or predict liver fibrosis in BA is mixed in the current literature.

In a study of 260 children with BA, an APRI > 1.22 was able to identify cirrhosis (at the time of presentation) with an AUROC of 0.83 (sensitivity 75% and specificity 84%).^[247] In a much smaller Korean study of 35 infants with BA, the AUROC of APRI to distinguish F3-4 was 0.92 and F4, 0.91 using optimal cut-points of 1.01 and 1.41, respectively,^[248] consistent with the thresholds proposed by Grieve et al.^[247,249] In a retrospective study of 91 infants with BA, METAVIR fibrosis was also significantly correlated with APRI ($R_s = 0.433$; $p < 0.05$).^[250] The mean APRI value was 0.76 in METAVIR F0-F1, 1.29 in F2-3, and 2.51 in F4 ($p < 0.001$). The AUROC of APRI for diagnosing F2-3 and F4 was 0.75 and 0.81, respectively. The APRI cutoff of 0.95 was 61% sensitive and 76% specific for F2-3, and a threshold of 1.66 was 71% sensitive and 83% specific for F4.

However, in another study of 29 patients with BA, APRI showed no significant correlations with METAVIR or Ishak global fibrosis scores.^[250] In a Chinese study of 24 children with BA (mean age 6.6 y) with prior Kasai portoenterostomy early in life undergoing liver biopsy, participants with METAVIR F0-2 had a median APRI and FIB-4 of 0.82 (vs. 1.9, $p = 0.053$) and 0.4 (vs. 0.22, $p = 0.49$), respectively, compared with F3-4.^[251] APRI had a positive correlation with fibrosis stage ($r = 0.583$) and showed significant differences between different fibrosis stages ($p = 0.035$), whereas FIB-4 did not. However, the AUROC of APRI for predicting F4 was only 0.56. Interestingly, in an Indian study of 48 children with neonatal cholestasis without BA, the mean APRI for METAVIR F0-3 was 1.38, whereas, for F4, it was 3.74. However, using an APRI threshold of 1.38, the AUROC to detect F4 among non-BA cholestatic infants was 0.75 with a sensitivity of 100% but a specificity of only 21.4%, thereby limiting its efficacy.

CFLD

CF is the most commonly inherited disease in Caucasian individuals manifesting in children. CFLD, with the development of portal hypertension, represents the third most common cause of death in CF, second only to pulmonary disease and lung transplant complications. Up to 7.5% of those with CF develop CFLD, and this typically becomes evident at a young age (median age 10.5 y). Liver biopsy is not essential to diagnose CFLD and thereby is not part of routine clinical care in the United States. However, a study comparing 51 Australian children with CFLD who underwent dual-pass liver biopsy with 104 age- and sex-matched children without CFLD demonstrated that APRI and FIB-4 not only identified those with CFLD but could provide information about severity of fibrosis.^[252] APRI had an AUROC of 0.8 for predicting advanced fibrosis, and a score > 0.462 indicated sevenfold increased odds of advanced fibrosis.

HBV

Cirrhosis in children with HBV is rare given that the majority of children are immunotolerant, although finding some degree of fibrosis (i.e., F2-3) in pediatric patients with HBV is not uncommon. In a Polish study of 71 children (age 4–17 y; mean age 10 y; mean ALT 83 IU/L) with biopsy-proven chronic HBV (HBeAg positive) and confirmed HBV DNA replication prior to antiviral treatment, 34 (48%) had advanced fibrosis. An APRI of > 0.59 differentiated children with significant fibrosis, with an AUROC of 0.75 PPV = 70% and NPV = 77%.^[253]

In a cohort study of 36 pediatric patients (up to age 20 y) with chronic HBV or HCV, the AUROC of APRI was 0.71 for identifying patients with any fibrosis (METAVIR classification) and 0.52 for identifying patients with cirrhosis.^[254] By disease, however, APRI had only modest performance characteristics when predicting fibrosis in patients with HBV and HCV (0.64 and 0.75, respectively) and in children age > 13 years old (0.65).

FibroTest-ActiTestTM has been validated in adults with chronic HCV infection as a noninvasive alternative to liver biopsy, but there are few data of its use in children with HBV. In a Scandinavian study of FibroTest in 25 children with HBV, there was no correlation between FibroTest scores and histological stage of fibrosis.^[255]

HCV

Cirrhosis is uncommon in children but has been reported. Studies examining the use of APRI or FIB-4 to assess fibrosis in children with HCV have been scarce. In an Egyptian study of 48 children with HCV, the AUROC curve for predicting significant fibrosis (F2-4 METAVIR) was 0.49 with APRI, which is not a clinically useful test.^[256]

In a prospective study of 50 Egyptian children with chronic HCV who had FibroTest measurements at the time of liver biopsy, the median FibroTest level increased linearly with advancing fibrosis stage. FibroTestTM values were 0.16 (0.07–0.25) in F0, 0.19 (0.18–0.24) in F1, 0.41 (0.20–0.66) in F2, 0.54 in F3, and 0.66 (0.43–0.77) in F4.^[257] A significant correlation was also found between individual FibroTestTM values and fibrosis stage, $r = 0.81$. At a FibroTestTM cutoff of 0.25, and the AUROC to differentiate F2-4 from F0-1 was 0.97 with 92% sensitivity and 96% specificity. Utilizing a higher FibroTestTM cutoff of 0.54, the AUROC was 0.92 to discriminate between F3-4 versus F0-2 with 71% sensitivity and 91% specificity.

There is also some limited evidence of discordance between FibroTestTM and METAVIR scores in children with HCV. In a small Polish study of 10 children with chronic HCV with FibroTestTM, there was no correlation of FibroTestTM values with advancing METAVIR fibrosis

staging.^[258] There was also discordance between FibroTest™ and METAVIR in 30% of cases, suggesting that FibroTest™ values correlate poorly with histopathological stage.

In conclusion, blood-based NILDA tests in children vary widely in their accuracy, even in detecting F3-4 fibrosis, and have difficulty discriminating earlier stages of fibrosis. These tests also have different disease-specific thresholds that correlate with histopathologic fibrosis and differ from adults. APRI and FIB-4 have been the most studied NILDA tests in children, but there is still insufficient evidence to recommend blood biomarkers as endpoints to monitor changes in fibrosis over time. Any blood-based NILDA that includes age (Table 5) should be used cautiously in children.

QUALITY OF EVIDENCE AND OTHER CONSIDERATIONS

Analyses supporting PICO 6 were based on very few studies and meta-analysis was not feasible. The quality of evidence was judged to be low for sensitivity and specificity estimates due to severe imprecision.

A simplified blood-based NILDA algorithm for detection of fibrosis and steatosis

In an effort to facilitate the incorporation of blood-based NILDA into clinical practice, the AASLD NILDA Writing Group developed an algorithm intended to be used by clinicians in need of a readily available and simple decision support tool (Figure 1). This algorithm was developed with the summary NILDA evidence highlighted earlier. We recommend that fibrosis staging begin with simple blood-based NILDA, including simple nonproprietary tests because of their wide availability and performance compared to proprietary tests, although these can be used where available. The left side of the algorithm aims to rule out advanced fibrosis.

Nonproprietary blood-based NILDA such as FIB-4 and NFS have sensitivities ranging from 60% to 75% for ruling out significant fibrosis and 75% to 85% for advanced fibrosis (depending on test cutoff and disease etiology) and the lowest negative likelihood ratios at proposed cutoff values across etiologies per our systematic review.^[94] Of the three major nonproprietary NILDA (FIB-4, APRI, and NFS in NAFLD), FIB-4 appears to have superior performance, particularly for the identification of F3-4 stages of fibrosis,^[94] which is the spectrum of fibrosis for which the tests were designed.^[42] NFS can be considered an equivalent to FIB-4 in patients with NAFLD in the assessment of advanced fibrosis.^[45] Thus, in the appropriate clinical setting (i.e., low pre-test probability), these tests should suffice to rule out significant/advanced fibrosis. A FIB-4 cutoff threshold of 1.3 has been proposed as accurate to rule out F3-4 in NAFLD patients,^[259] and our systematic review indicated a higher sensitivity, as expected for the lower FIB-4 cutoff 1.3, but higher DOR for the standard 1.45 threshold.^[94] Confirmatory testing such as imaging-based NILDA should be performed for patients with values between the lower and upper thresholds. For those with blood-based values above the threshold for advanced fibrosis, imaging-based NILDA can be considered for confirmation and patients should be referred for HCC surveillance per AASLD guidelines.^[260] These thresholds correspond to the highly specific cutoff values validated for the recognition of advanced fibrosis (FIB-4 and NFS, specificity of 91% to 97%) across etiologies (except for NFS, which is only for NAFLD) per our systematic review;^[94] a revised upper FIB-4 cutoff value of 2.67 has been proposed to rule in F3-4 in NAFLD,^[68] and although our systematic review indicated a lower DOR for the standard upper FIB-4 threshold of 3.25, both cutoff values had similar high specificity of 94% to “rule-in” advanced fibrosis in NAFLD patients.^[94] Although imaging-based NILDA are more accurate than blood-based NILDA in some situations, elastography methods are not as widely available. As imaging-based NILDA become more readily available in practice, their sequential incorporation with blood-based NILDA in

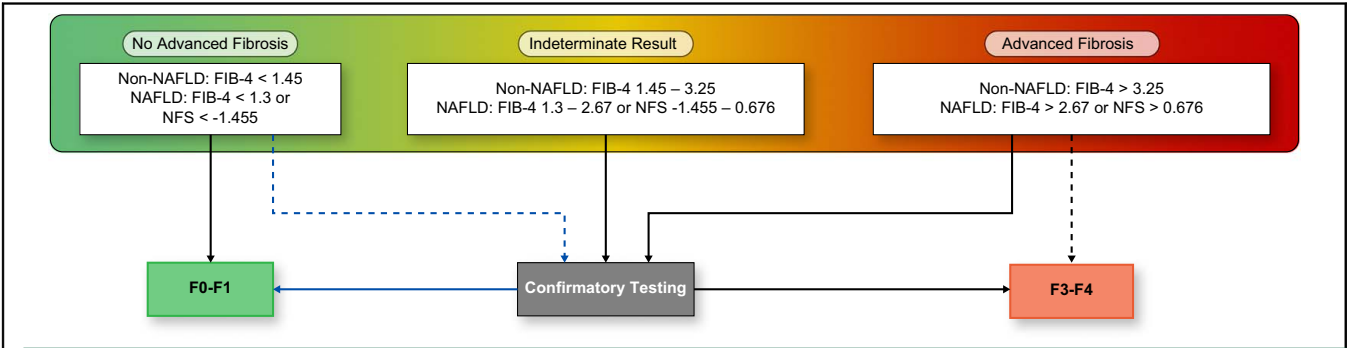


FIGURE 1 Simplified Blood-based NILDA algorithm for the clinician. Note: See also AASLD Guidelines on imaging-based NILDA (Non-Invasive Liver Disease Assessment). Abbreviations: F, fibrosis; FIB-4, fibrosis 4 index; NAFLD, nonalcoholic fatty liver disease; NFS, NAFLD fibrosis score.

TABLE 11 Blood-Based NILDA: major areas for future research

Comparative studies of proprietary and nonproprietary blood-based NILDA are needed in the primary care population, with lower expected prevalence of advanced fibrosis and with attention to cost-effectiveness to generalize the application of NILDA.
Studies on NILDA should include diverse populations and children.
All findings among patients with NAFLD in this guideline will need to be confirmed among patients with the new MASLD and SLD nomenclature.
Confirmation that novel markers such as PRO-C3, a serologic biomarker that detects formation of type III collagen from activated myofibroblasts, especially when combined with age, presence of T2DM, and platelet count, are superior to APRI, and FIB-4 in MASLD and NASH is needed.
Emerging data with newer biomarkers such as ELF™ may improve the accuracy of blood-based NILDA in NAFLD and MASLD.
Comparative studies combining both blood-based and imaging-based tests synchronously and sequentially are needed to reflect clinical practice, with recognition of test utility by insurance and third-party payors.
Blood-based algorithms have the potential to help identify those with steatosis, but, to enhance clinical utility, they need to differentiate simple steatosis from MASLD and NASH.
Utilization of artificial intelligence and machine-learning tools should allow for incorporation of demographics and a wide array of clinical data to improve diagnosis and management of CLD.
Longitudinal studies of NILDA to assess the natural history of chronic liver diseases, clinical outcomes, and changes with therapy are needed.

Abbreviations: APRI, AST-to-platelet ratio index; AST, aspartate amino-transferase; CLD, chronic liver disease; ELF, enhanced liver fibrosis; FIB-4, Fibrosis-4 Index; MASLD, metabolic dysfunction-associated steatotic liver disease; NASH, nonalcoholic steatohepatitis; NILDA, noninvasive liver disease assessments; PRO-C3: N-terminal propeptide of type III collagen; SLD, steatotic liver disease; T2DM, type II diabetes mellitus.

clinical decision-making is expected to grow. Whenever more granularity is needed (i.e., start of antiviral treatment for a patient with HBV and significant fibrosis, initiating HCC surveillance), clinicians should refer to the associated NILDA Systematic Reviews that have more detail on NILDA^[4,6,94] or specific guidance documents.^[3,5] Per our systematic review, blood-based NILDA for steatosis are not accurate enough for daily practice,^[94] and the AASLD NILDA Writing Committee recommends utilizing imaging-based NILDA for the identification of steatotic liver disease.^[3]

Summary

NILDA has replaced liver biopsy in clinical practice in many situations. Because of the rapid evolution of the field and predetermined requirements for studies to be

incorporated in our systematic reviews, we were not able to include every published study on the topic; in particular, studies with smaller sample sizes, those that did not have liver histology to assess fibrosis or, for fatty liver, did not have histology/MRS/MR-PDFF as the reference standard. Many studies with mixed etiologies or overlapping diseases were excluded. In blood-based NILDA with upper and lower thresholds to rule in or out fibrosis severity, up to one-third of patients can have indeterminate ranges that require additional diagnostic tests such as imaging-based NILDA (see *AASLD Practice Guideline: Imaging-Based Noninvasive Liver Disease Assessments [NILDA] of Hepatic Fibrosis and Steatosis*).^[3]

FUTURE RESEARCH

Although substantial progress has been made in the area of NILDA, there are still many opportunities for future research. In the era of precision medicine, high-throughput technologies applied to experimental models will continue to generate a wealth of novel disease and injury-specific blood-based biomarkers for dynamic fibrosis assessment. Selection and validation of candidate biomarkers for fibrosis assessment from these multi-omics databases will be challenging. Progress in this field requires a paradigm shift from using a static and semi-quantitative assessment of fibrosis as the reference standard, towards developing dynamic disease-specific models of clinical relevance that are associated with outcomes. Our writing group identified several major areas for future research that are needed, as detailed in [Table 11](#).

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CONFLICTS OF INTEREST

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